Programming and Diagnostic Utility (PDU)

Table of Contents

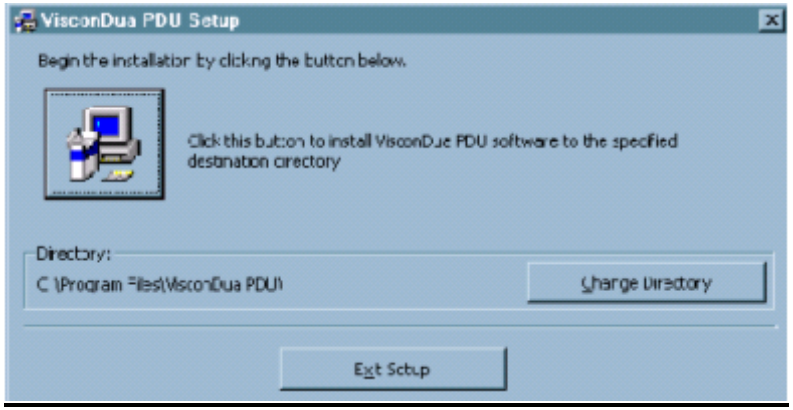
Introduction to PDU 2
PDU Installation 3
How to connect PDU to the RTU 4
Windows in PDU 6
Windows in PDU 7
Configurator window 10
Configurator window: Start 11
Configurator window: Scan for RTU 12
Configurator window: Getting Started tap window: Protocol Setting on Getting Started window 13
Configurator window: Abbreviation tap window: Abbreviation 14
Configurator window: Topology 15
Configurator window: Notes 16
Configurator window: Protocol 17
Configurator window: Command 18
Configurator window: Time Synchronization 19
Comms. Window: Show menu 20
Comms. Window: Option menu 21
Comms. Window: Establish Communication Between PDU and RTU 22
Comms. Window: Change the Comport Configuration 24
Comms. Window: Change the Configuration Option 25
Comms. Window: Activity Tap Window 26
Comms. Window: Seek Tap window 27
Configurator window: Counter 28
Configurator window: System Setting 29
Configurator window: System Setting 30
Configurator window: Address and Comms 31
Configurator window: Parameters 32
Configurator window: IO Setting for Digital Input 33
Configurator window: IO Setting for Digital Input 35
Configurator window: IO Setting for Digital Output 36
Configurator window: Digital Setting for Digital Input 37
Configurator window:IO Setting for Analog Input 37
Configurator window: IO Setting for Analog Input 37
Configurator window: IO Setting System Point 38

Introduction to PDU

- PDU is **P**rogramming and **D**iagnostic **U**tility.
- Used for installation and maintenance of RTU.
- Used IEC 60870-5-101 standard protocol.
- User-friendly software with windows-based environment.
- Runs on Windows 95 or above.
- Runs on personal computer with minimum requirement of 386 processor.
- Requires a basic PC knowledge under Windows environment.

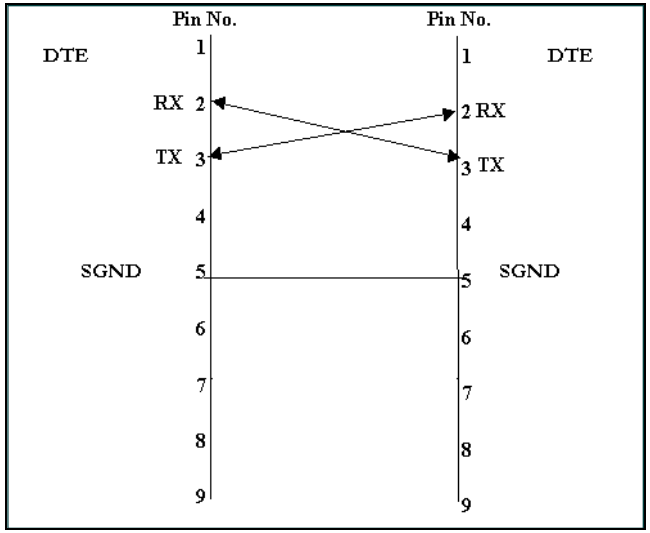
PDU Installation

- Insert PDU installation CD.
- Run SETUP.EXE file.
- By default the Viscon Dua-PDU Setup will install the application in the Program File.
- To change to other directory, click Select Directory button and enter new directory.
- Click the Installation icon to proceed or click Exit Setup to cancel.



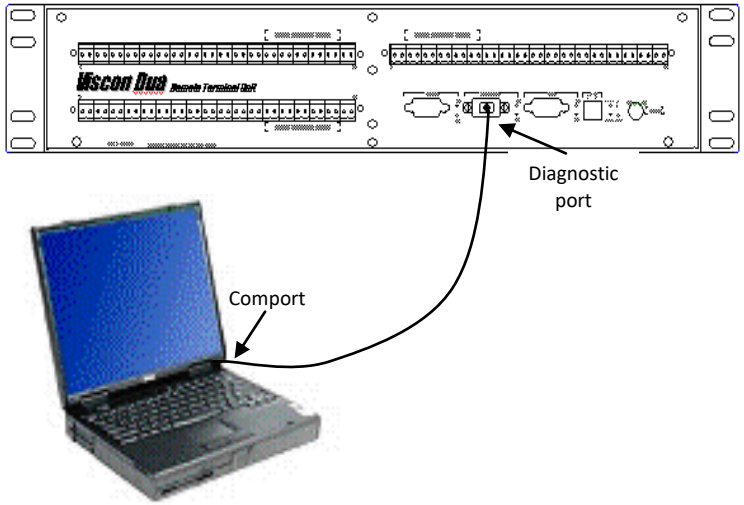
Pin Assignment for PDU to the RTU Connection

- Prepare a three-core cable and a pair of DB9 male and female port connector.
- Connect DB9 port connectors at each end of the cable. Refer pin assignment.



How to connect PDU to the RTU

- Connect the DB9 female connector to the Comport of the PC/laptop that runs Viscon Dua PDU
- Connect the other end of DB9 male connector to the Diagnostic port of Viscon Dua RTU Input/ Output module.
- Click PDU icon on PC/laptop



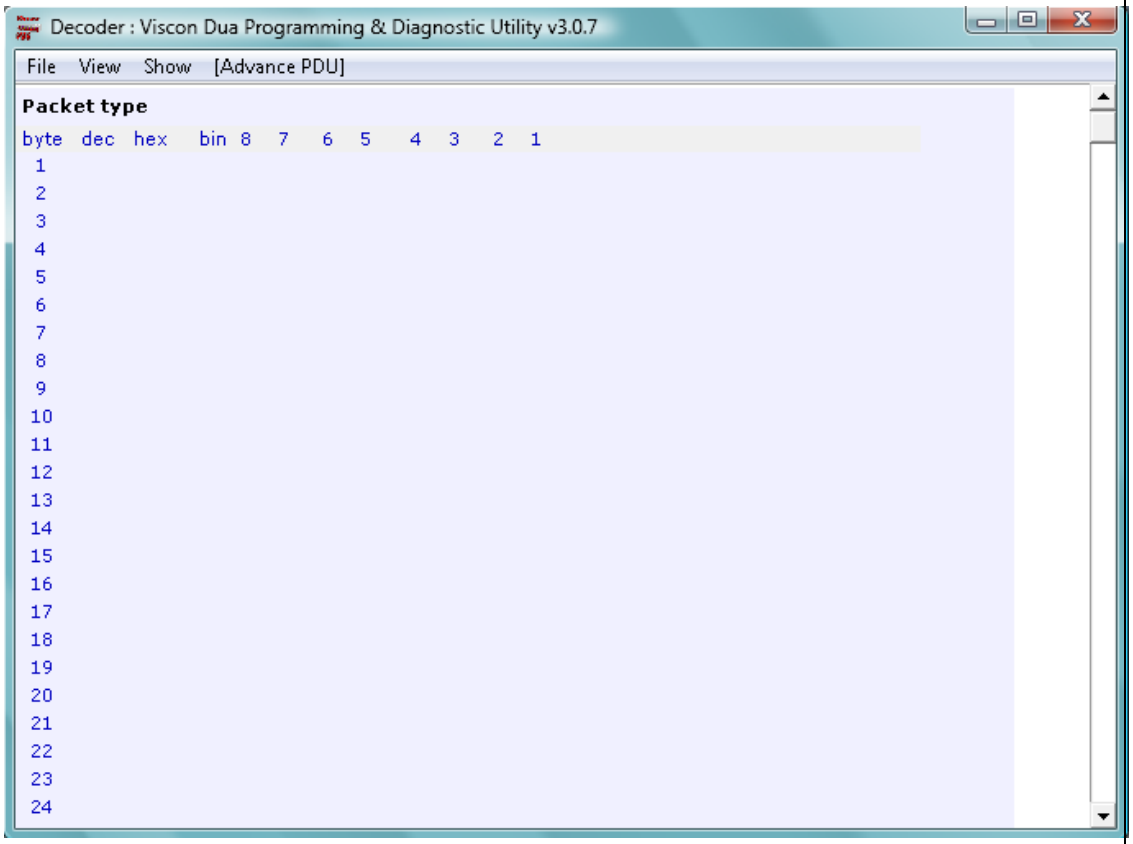
Windows in PDU

- There are 4 windows inside PDU which are:
  - Configurator window
  - Communication window
  - Translator window
  - Decoder window
- The functions of **Configurator** window:
  - to set the configuration of the RTU
  - simulate MS application to interact with RTU

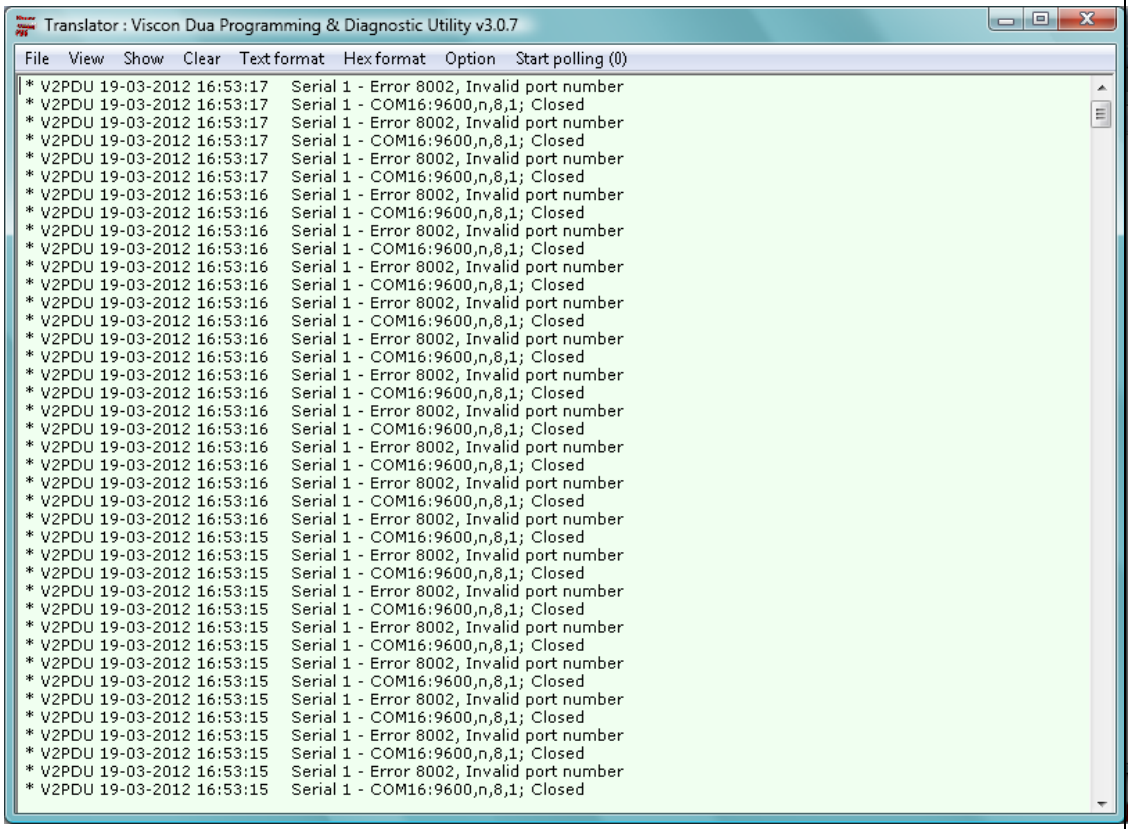


Windows in PDU

- The function of **Decoder** window:
  - Display Tx and RX packet in protocol format

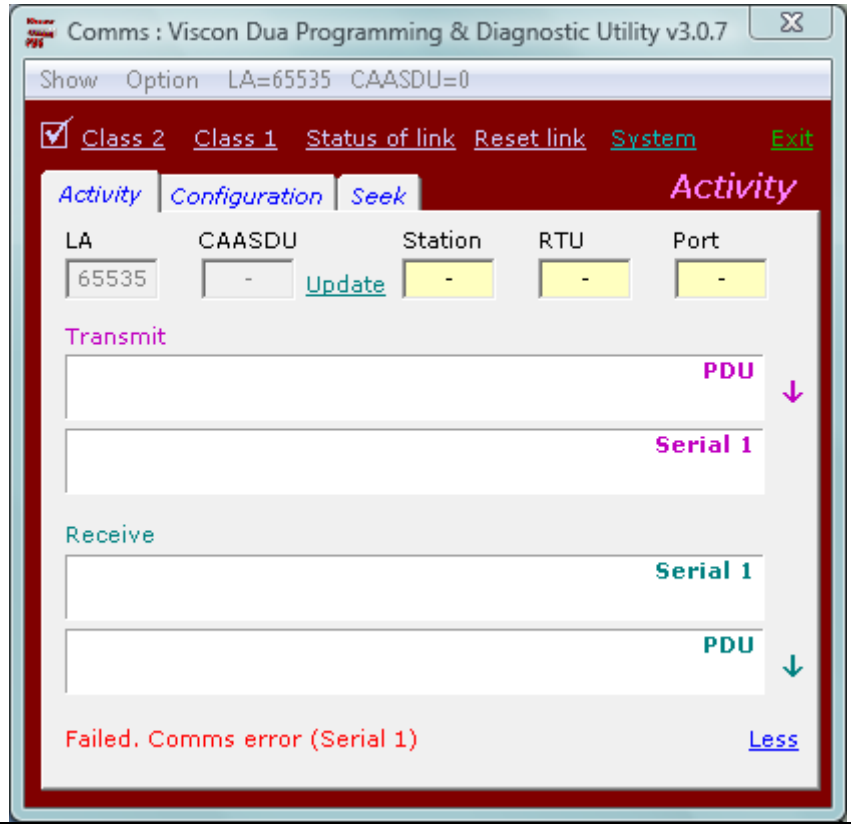


- The functions of **Translator** window:
  - Translate TX and RX packet into simple English



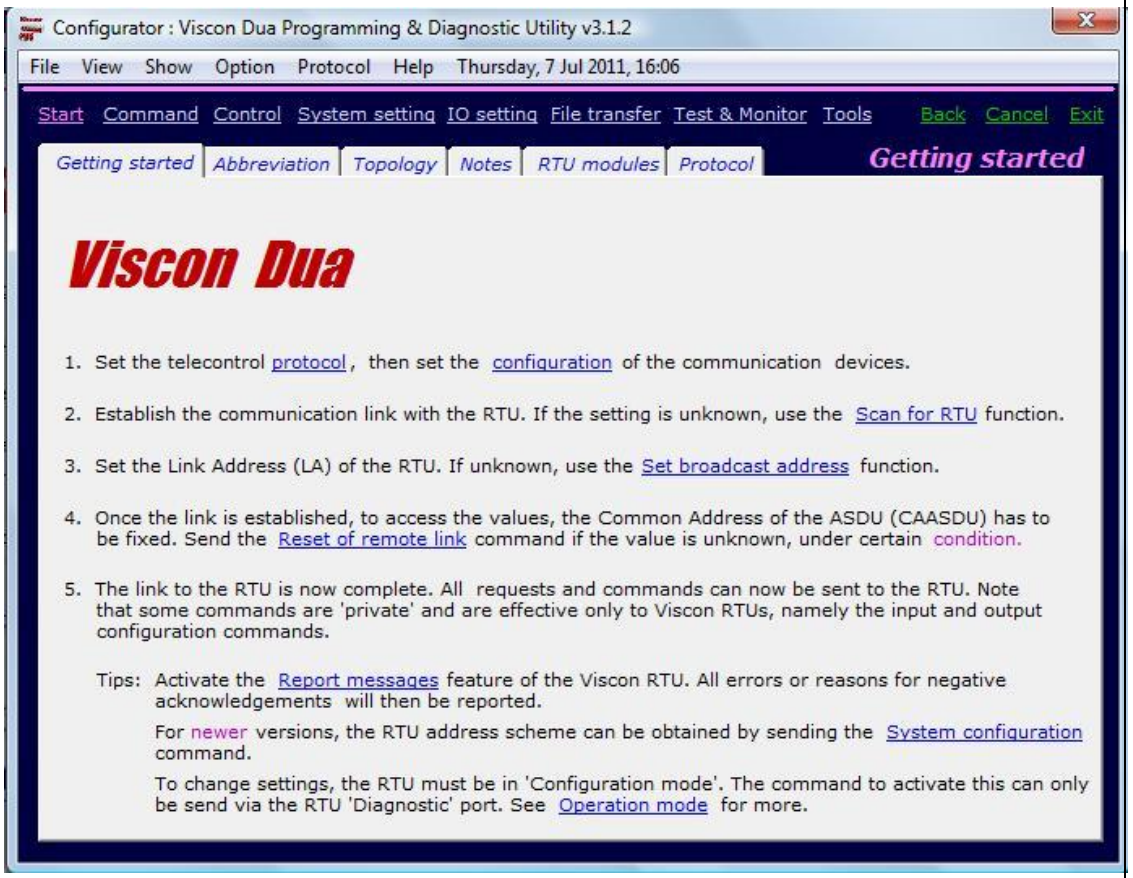
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- The function of **Comms**. Window:
  - Set the communication configuration and show communication status



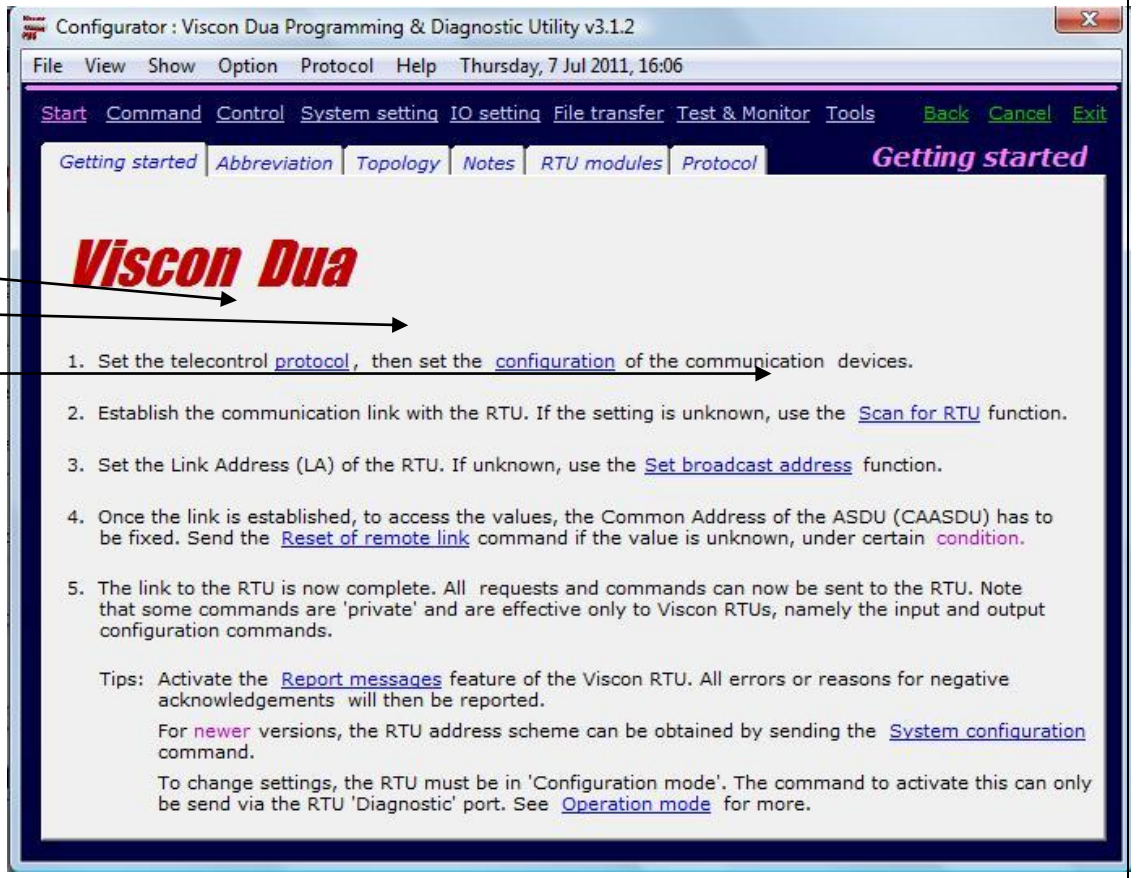
Configurator window

- Components on Configurator window:
  - Poll down menu
  - Command
  - Tap window



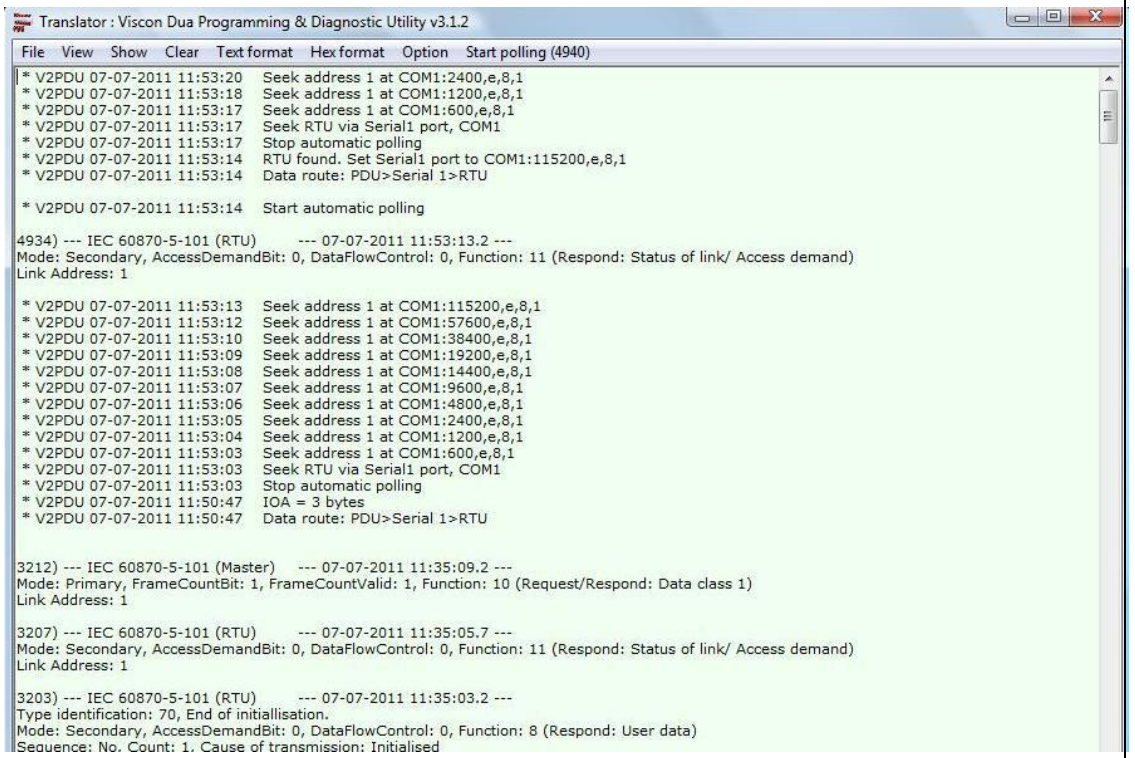
Configurator window: Start

- Default display upon start-up is **Getting Started** window.
- Important commands on **Getting Started** window:
  - Protocol
  - Configuration
  - Scan for RTU
- These commands will refer to communication configurations on **Comms. window**.
- Click these commands in sequence to establish communication between PDU and RTU.
- If communication successfully establish, new events will be recorded on **Translator window**.



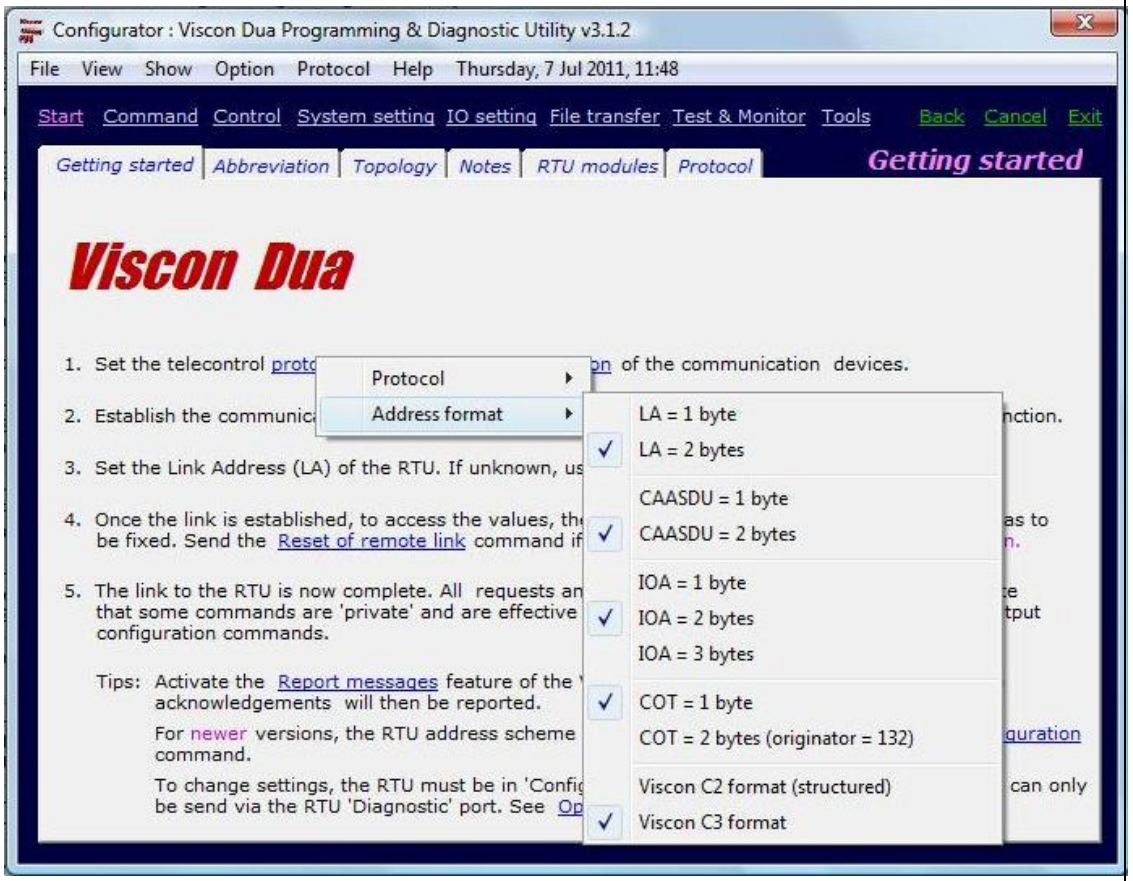
Configurator window: Scan for RTU

- Events shown on Translator window when **Scan for RTU** command is executed.



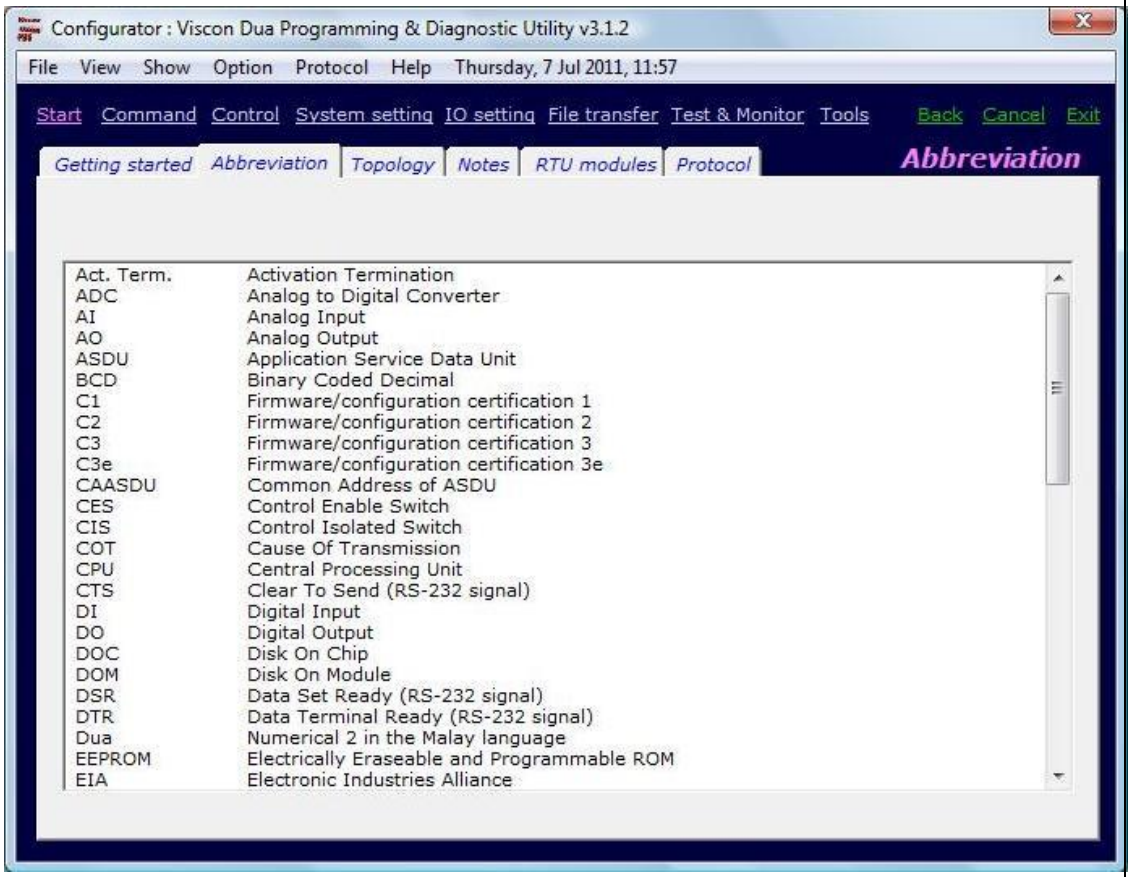
Configurator window: Getting Started tap window: Protocol Setting on Getting Started window

- Address format is set according to RTU configuration



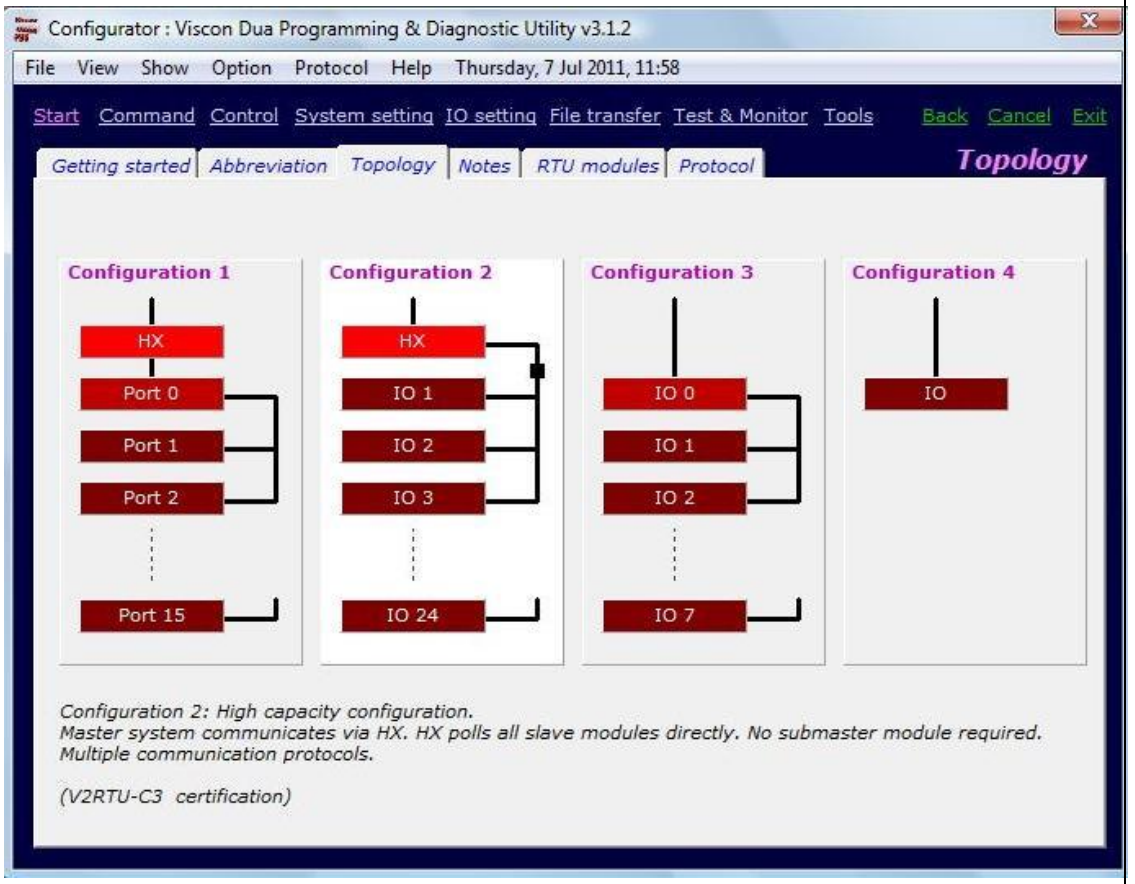
Configurator window: Abbreviation tap window: Abbreviation

- Scroll down the window to know the abbreviation used within the operation of PDU.



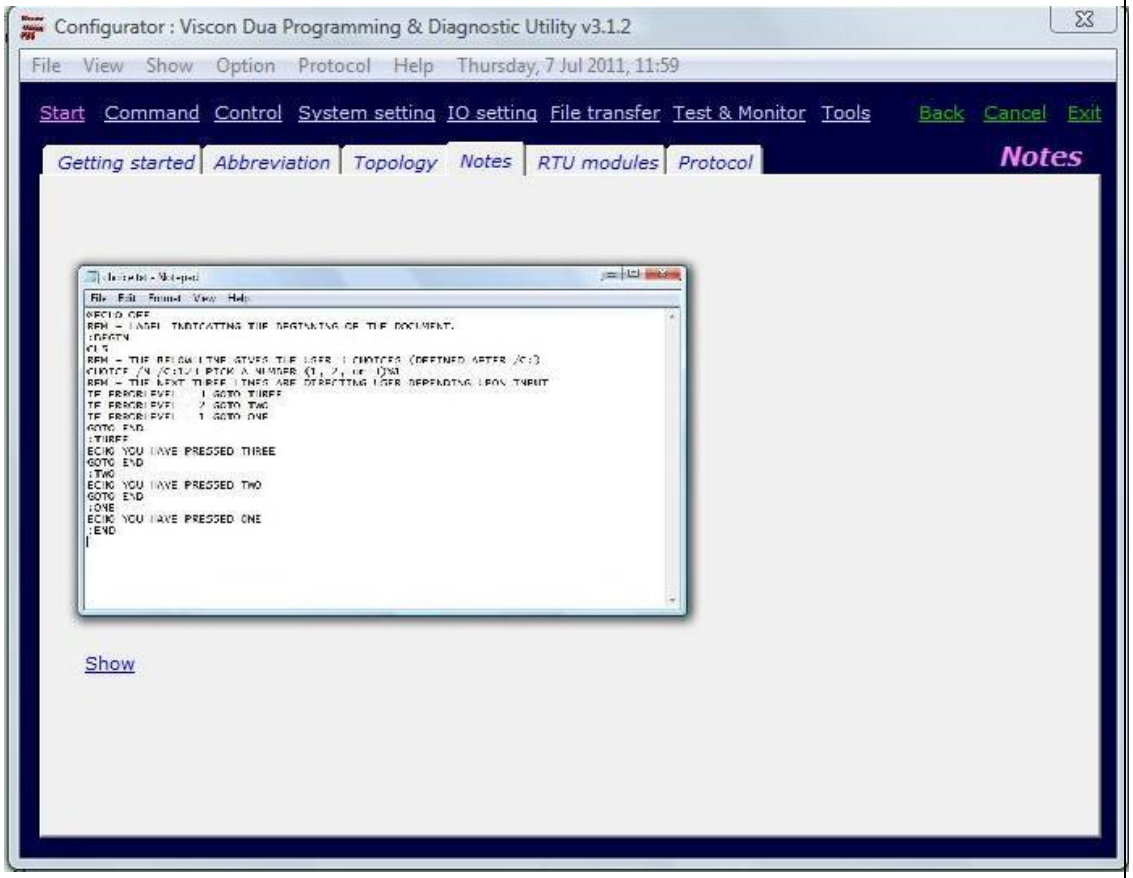
Configurator window: Topology

- Display options of RTU configuration supported by the PDU.



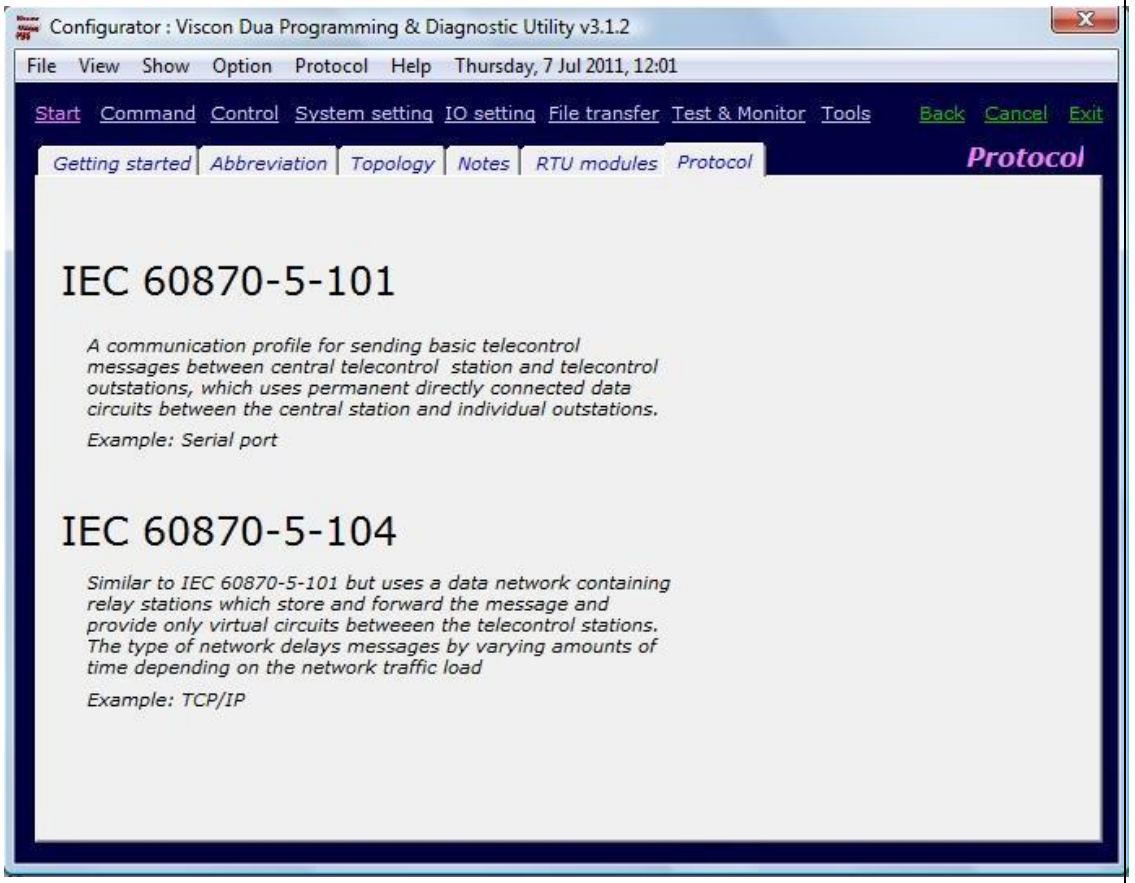
Configurator window: Notes

- Allow user to create a note in .txt file.
- Click **Show** to start editing a new notes.



Configurator window: Protocol

- Description on IEC protocol supported by the PDU.



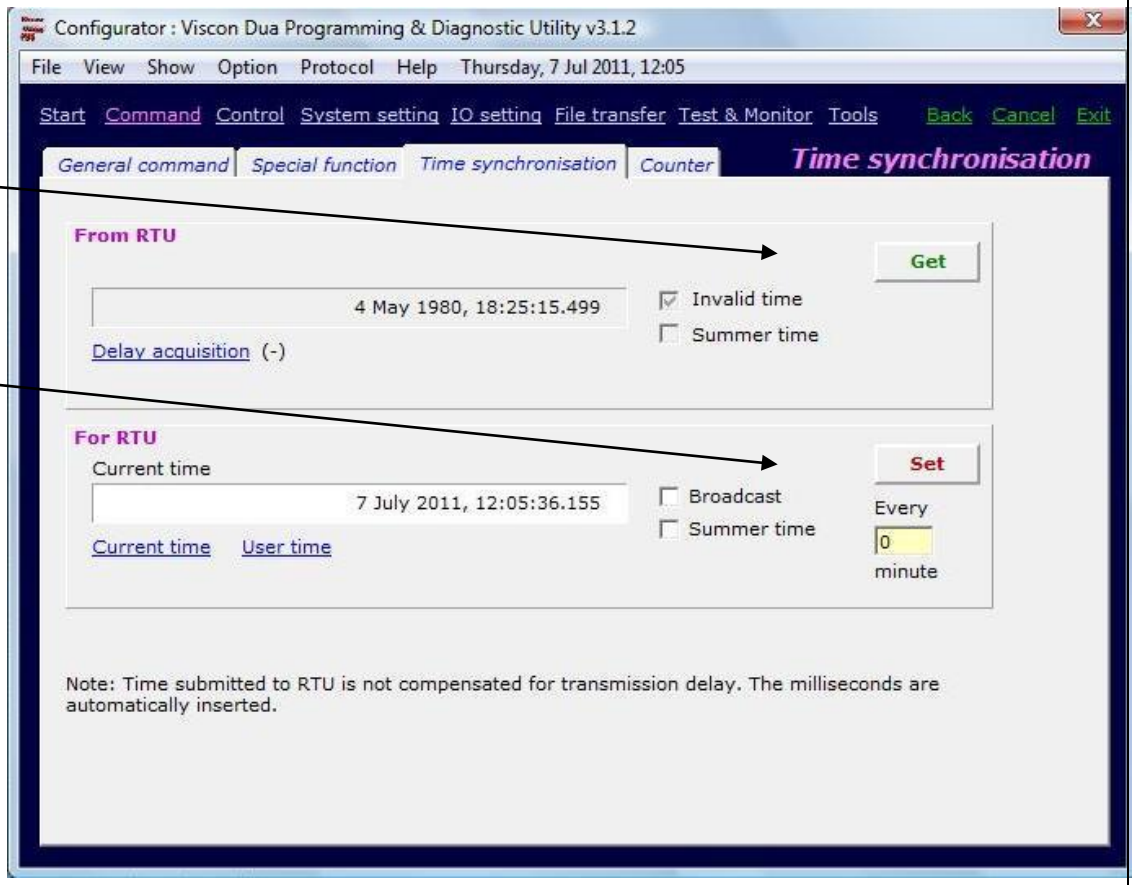
Configurator window: Command

- General commands are broadcast commands to the RTU.



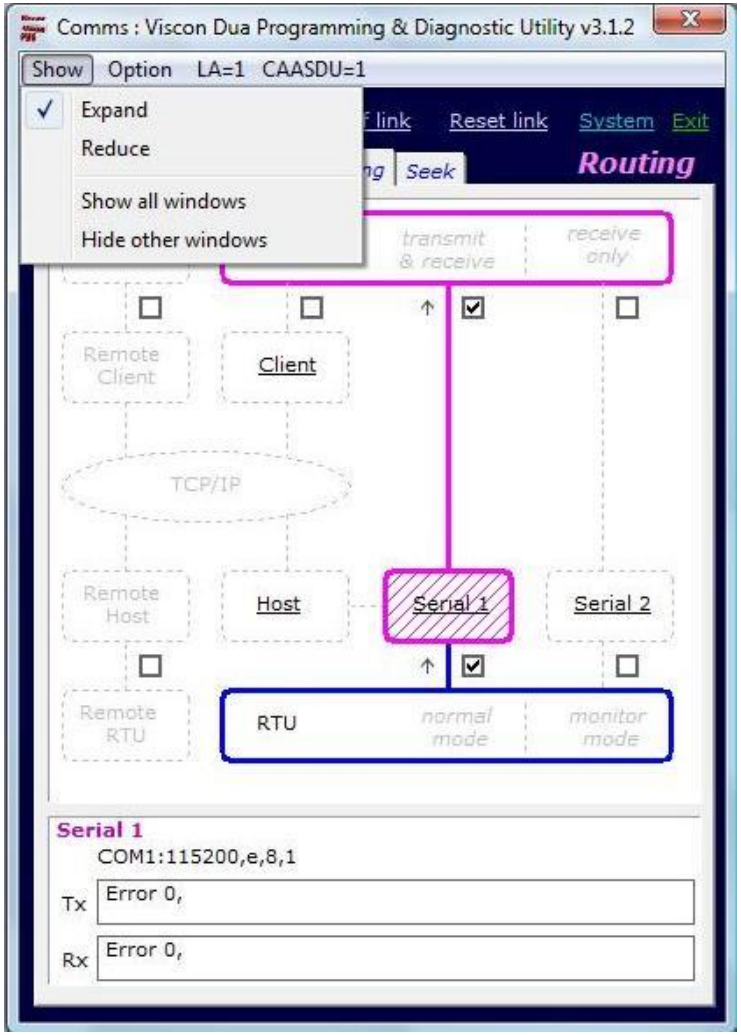
Configurator window: Time Synchronization

- Get current RTU time from RTU.
  - Click **Get** button to get time from RTU.
- Set time from PC (PDU) to RTU.
  - Type in new time of get current time from PC and click **Set** button to send new time to RTU.



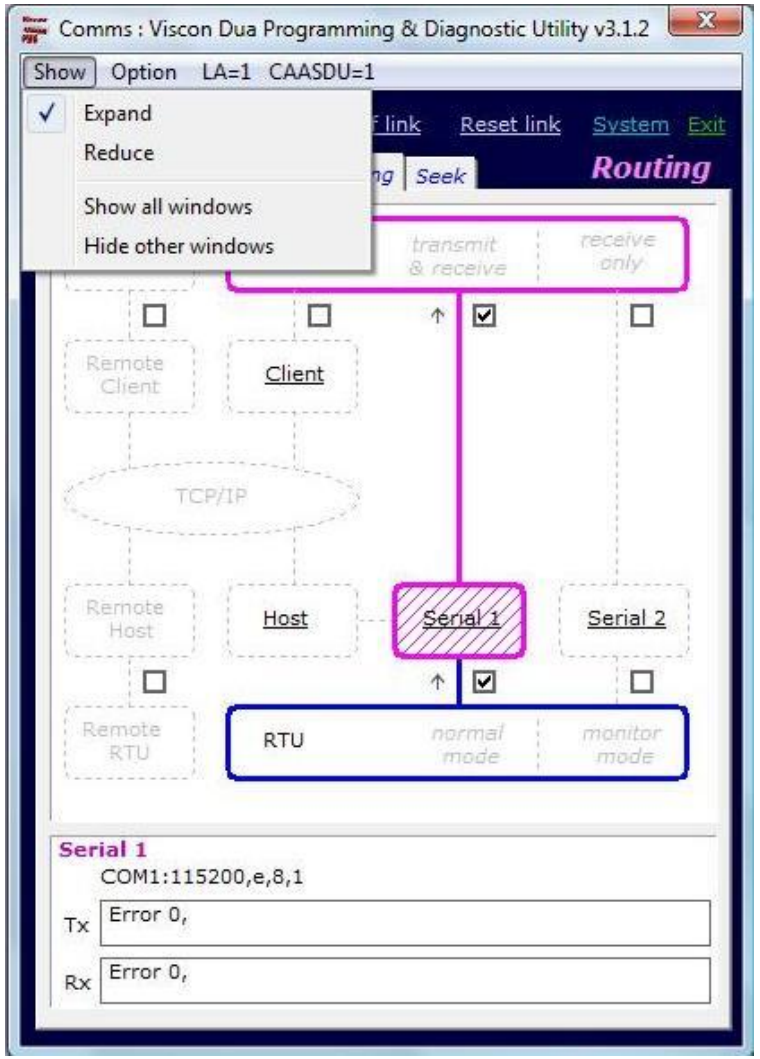
Comms. Window

- Components on Comms window:
  - Poll down menu
  - Command
  - Tap window
- Components on Comms window:
- Two poll down menu which are:
  - Show menu
  - Option menu



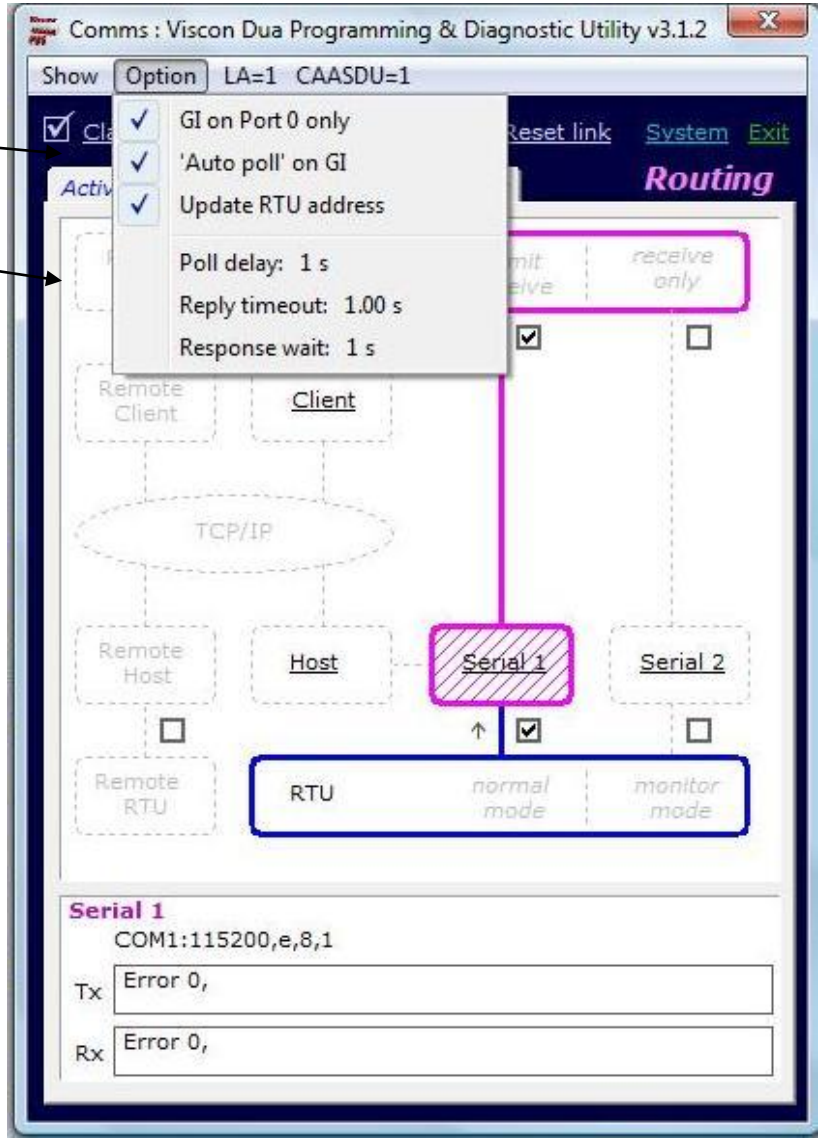
Comms. Window: Show menu

- Commands under Show menu:
  - Expend: to expand window to show all information
  - Reduce: to reduce window to show only Tx/Rx packet window
  - Show all windows: to display all 4 windows in PDU
  - Hide other windows: to hide other window except Comms. window



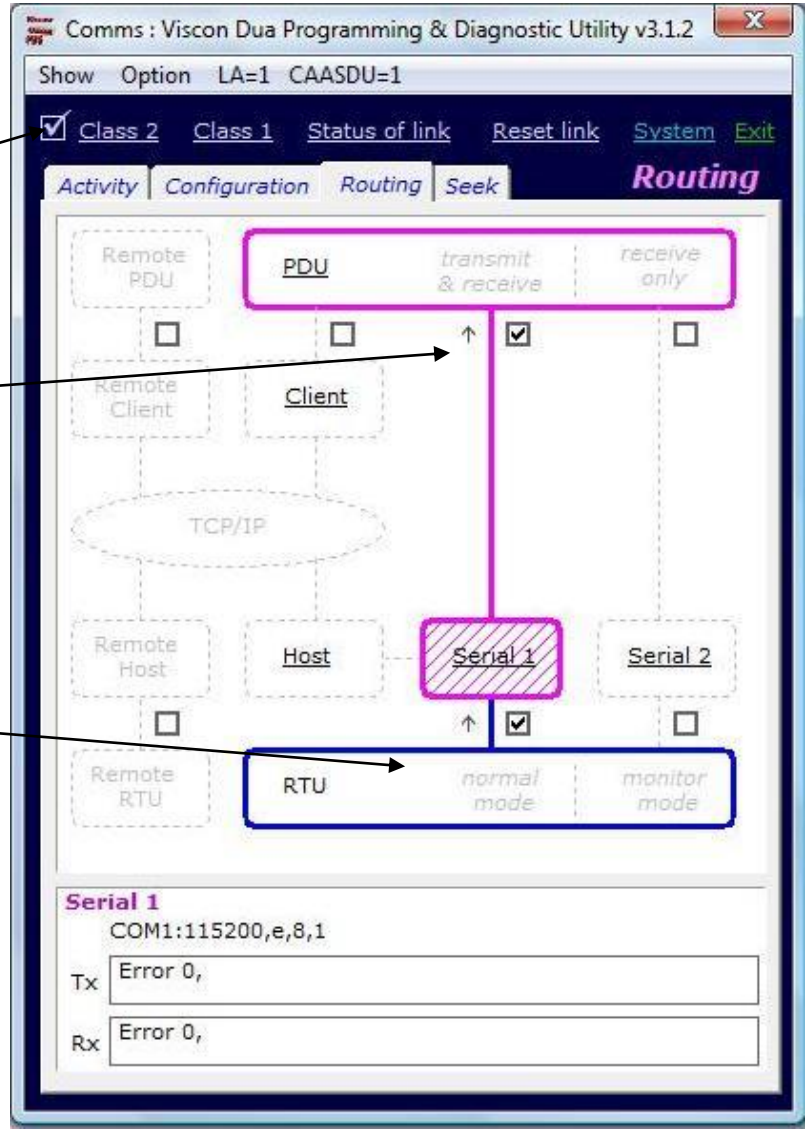
Comms Window: Option window

- Commands under **Option** menu:
  - **Tick option to select**
  - **Type a new value**



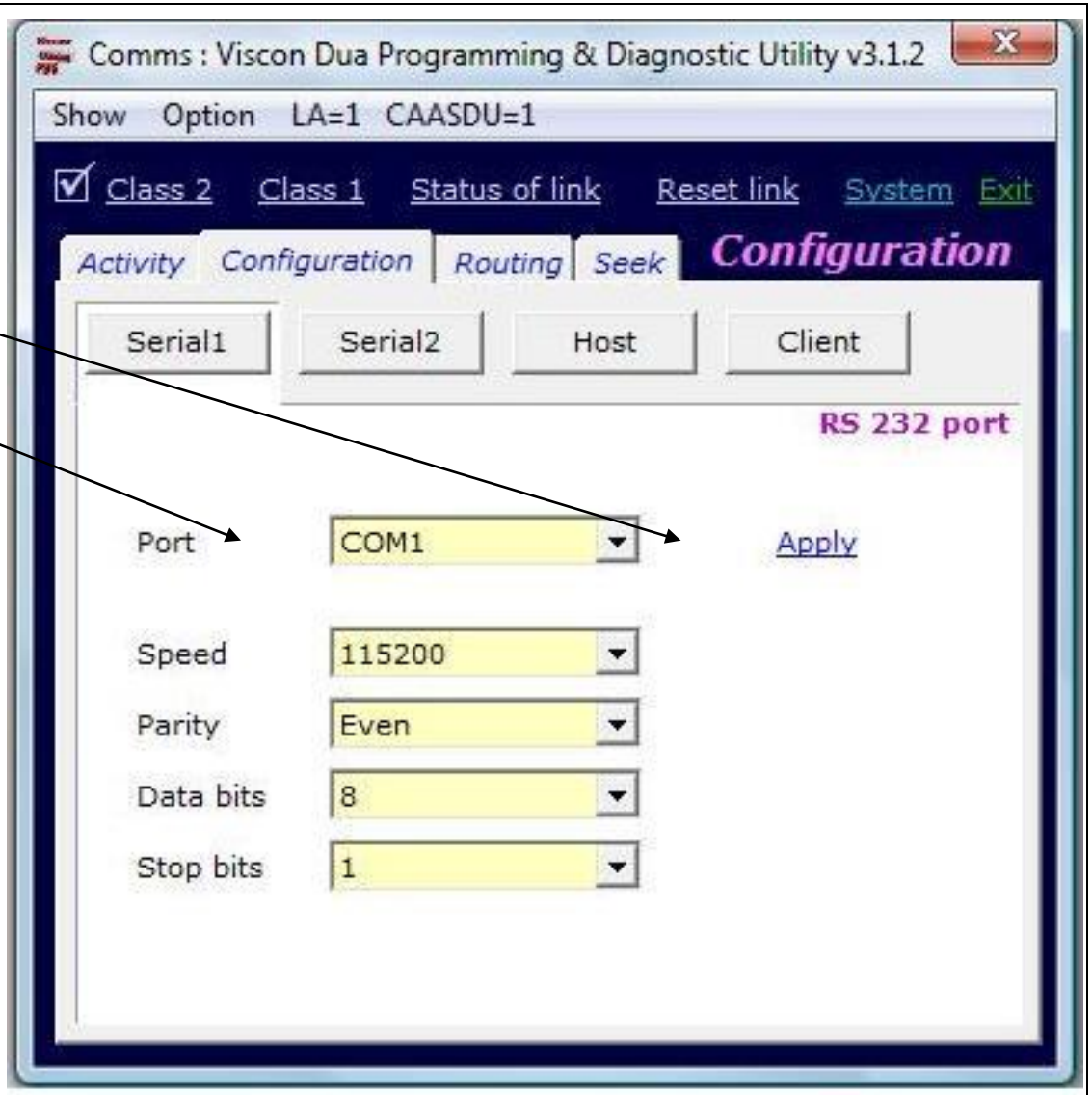
Comms. Window: Establish Communication Between PDU and RTU

1. Go to **Routing** tap window. It shows 4 types of routing.
2. The diagram shows the interconnection between PDU and RTU by using Serial 1.
3. Tick small box beside **Class 2** to poll Class 2 continuously.
4. Tick on the small box to select Serial 1.
5. The communication between PDU and RTU establishes when the background color changes from red to blue.
6. The '↑' will show the data flow between PDU and RTU.



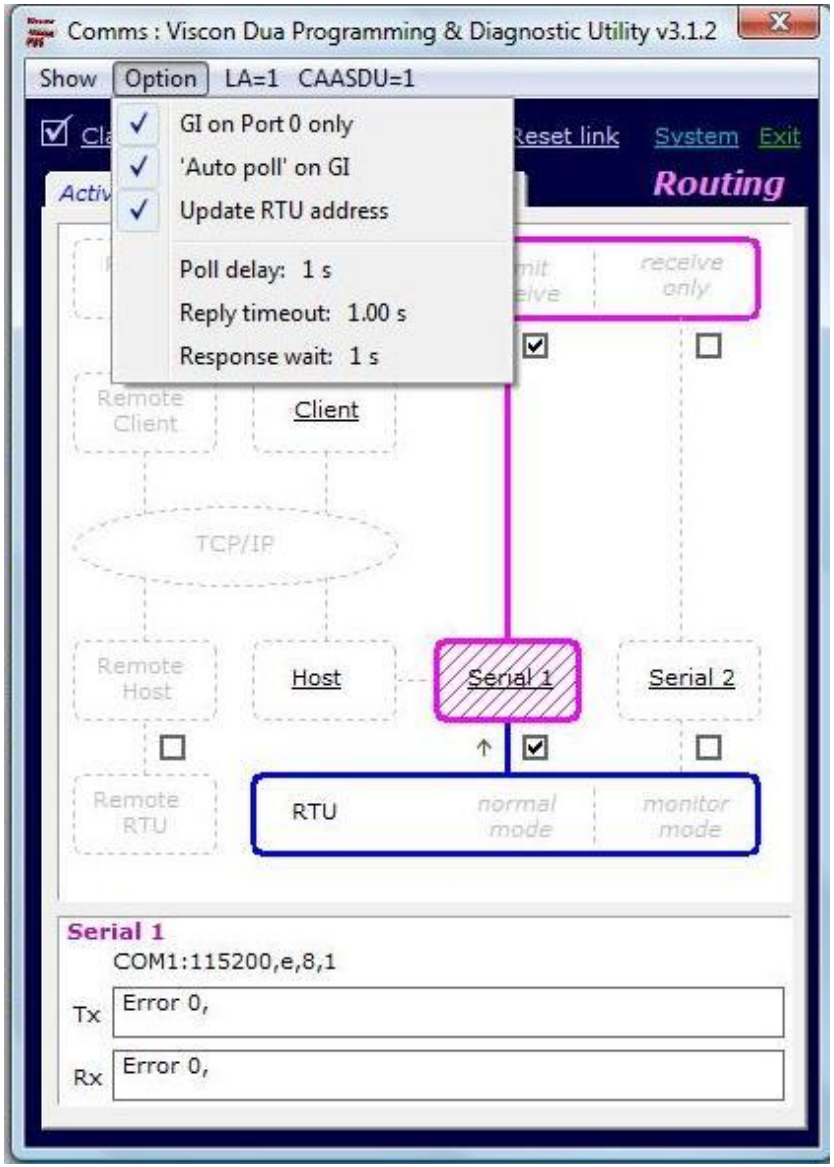
Comms. Window: Change the Comport Configuration

1. Go to **Configuration** tap window.
2. Choose port = COM1 and set the configuration as shown.
3. Click Apply to accept the setting.



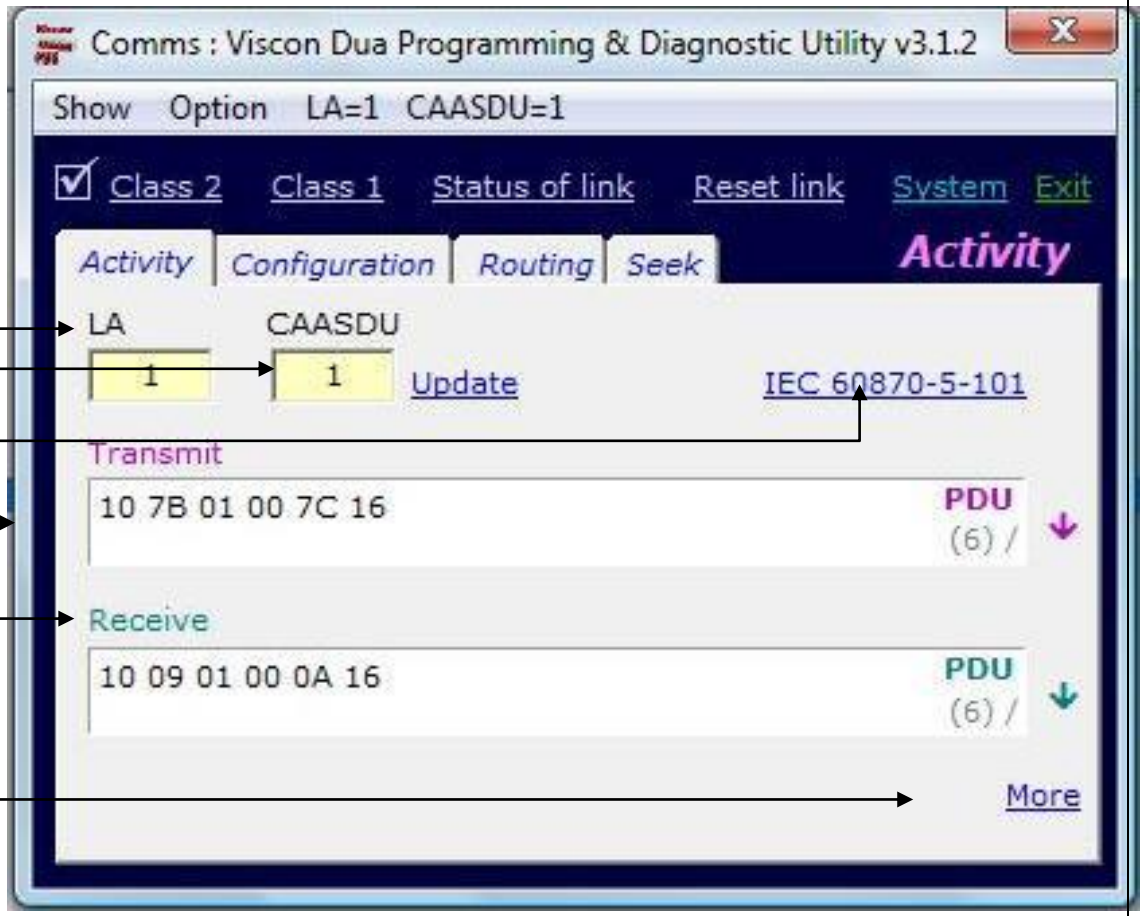
Comms. Window: Change the Configuration Option

1. Go to **Option** menu.
2. Tick option for '**GI'**, '**Auto Poll**' and '**Update RTU Address**'
3. Set new value for **Poll delay**, **Reply timeout** and **Respond wait**.



Comms. Window: Activity Tap Window

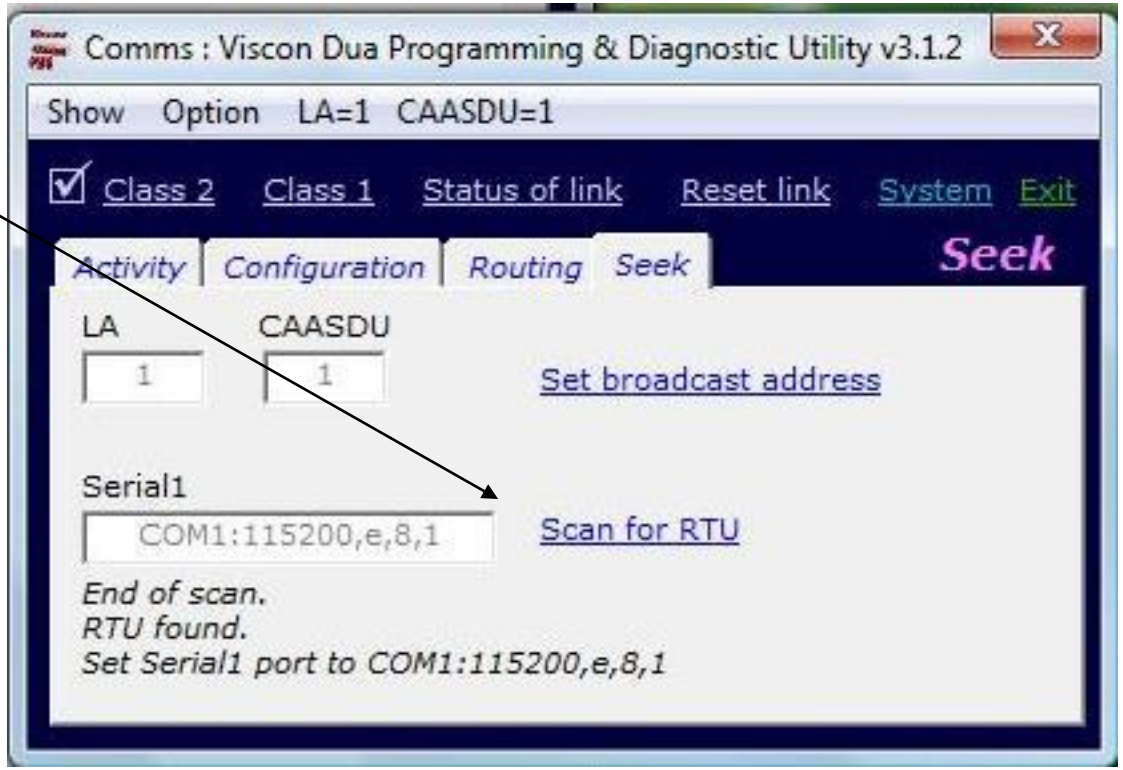
- Link address number (LA)
- CAASDU number
- Standard protocol used for communication
- Transmit packet from PDU to RTU
- Receive packet from RTU to PDU
- Click **More** to expand the window



Comms. Window: Seek Tap window

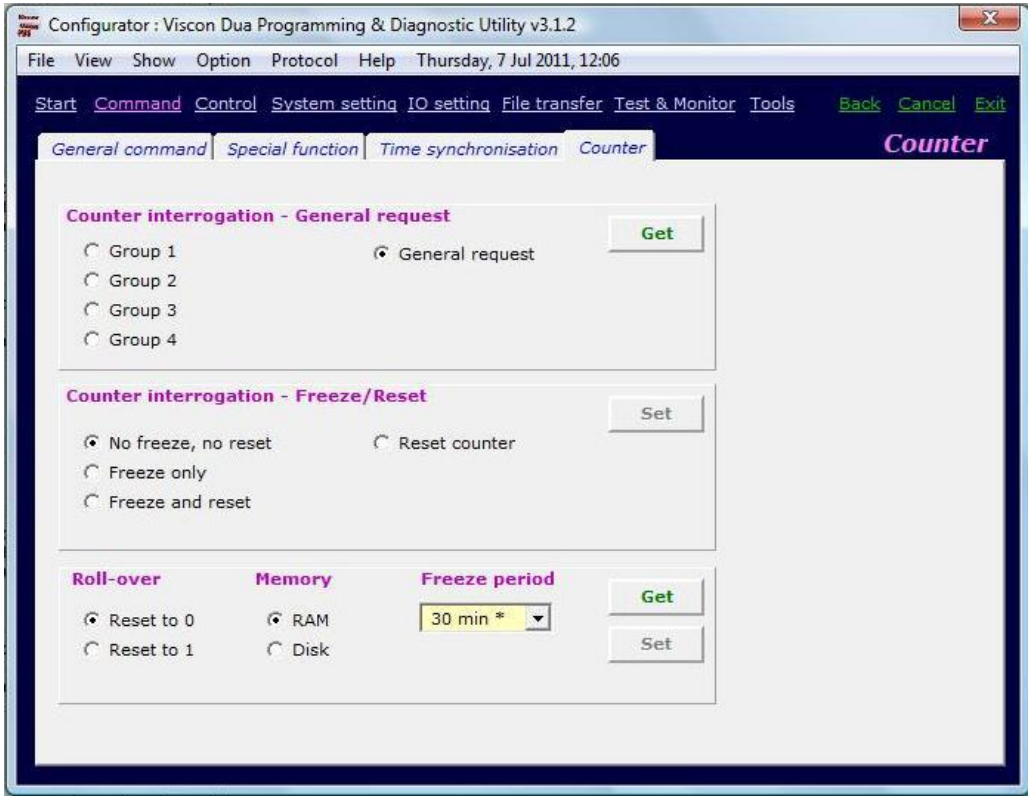
1. If communication fails to establish, go to Seek tap window and click **Scan for RTU.**

2. PDU will seek for RTU manually.



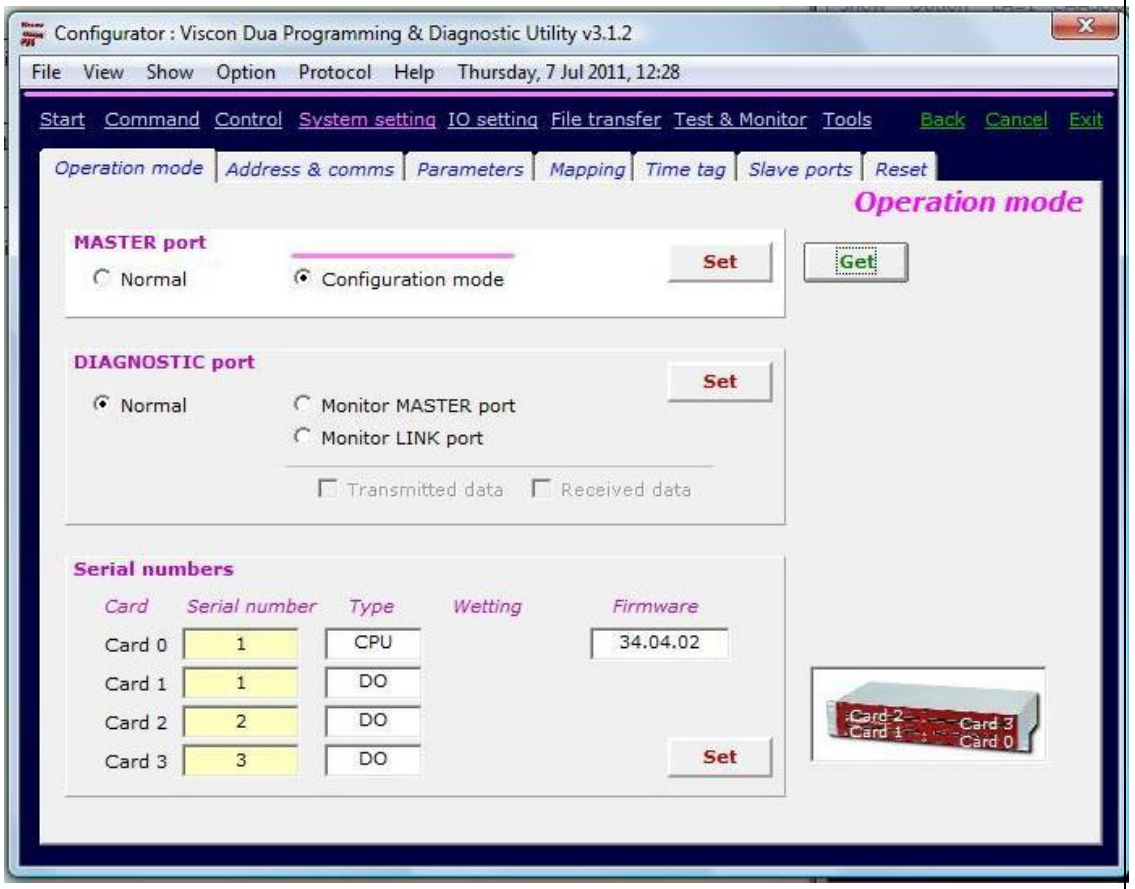
Configurator window: Counter

- Setting for DI Counter points and how they should behave during operation.



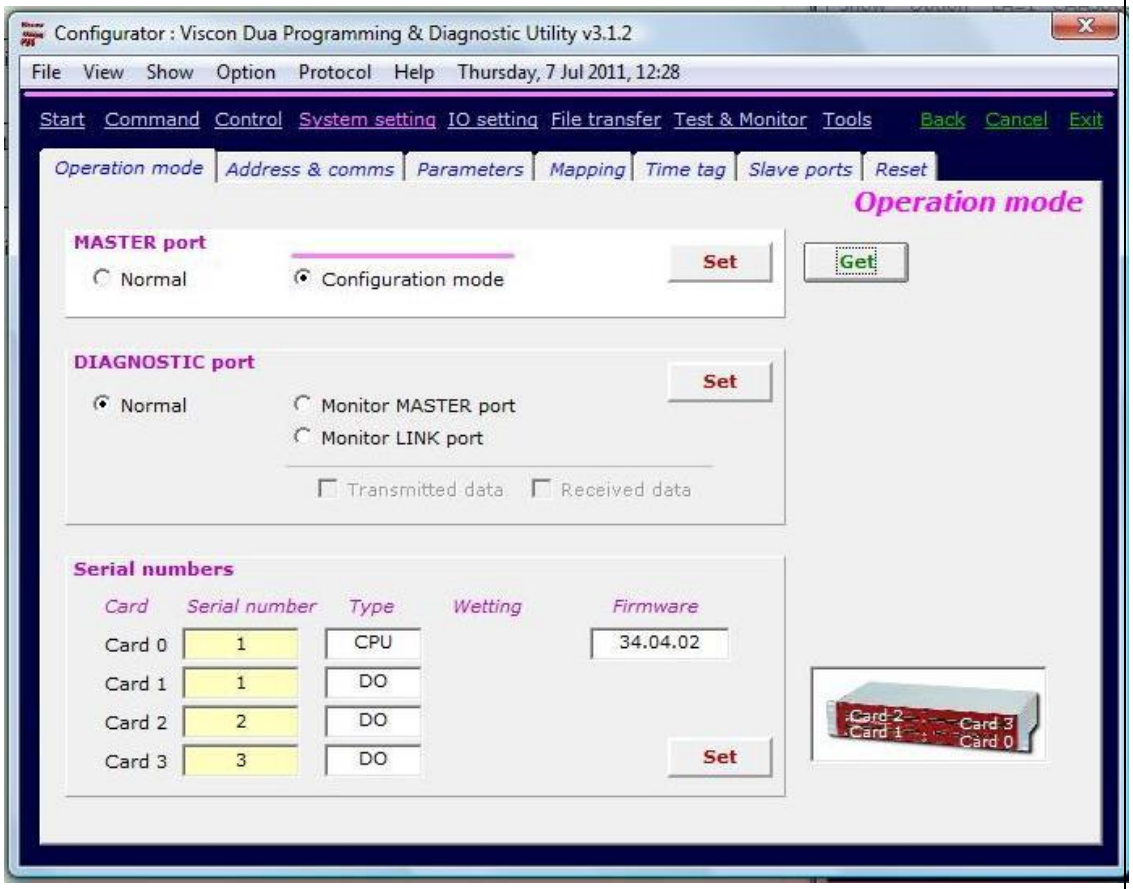
Configurator window: System Setting

- MASTER port:
  - Set to **Normal** if only allow PDU to poll information from RTU. Allows only Get command.
  - Set to **Configuration Mode** to allow PDU to do configuration as well as poll information from RTU.
- DIAGNOSTIC port:
  - Set to **Normal** if only allow PDU to poll information from RTU. Allows only Get command.
  - Set to **Monitor MASTER port** to record activity at MASTER port ('T' line).
  - Set to **Monitor LINK port** to record activity at LINK port (inter-port link).
  - Option to record Tx or Rx packet only.



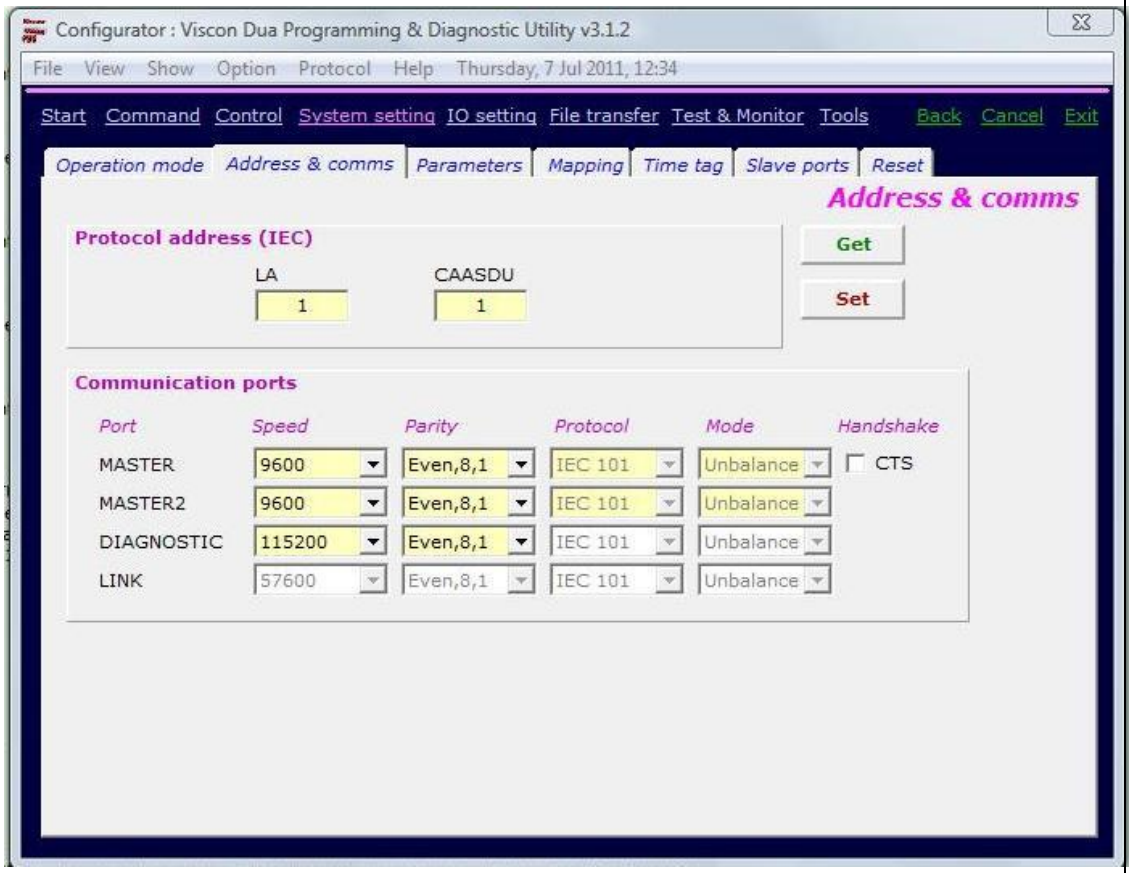
Configurator window: System Setting

- Serial Numbers
  - To record information on RTU IO card Serial No. and firmware used.



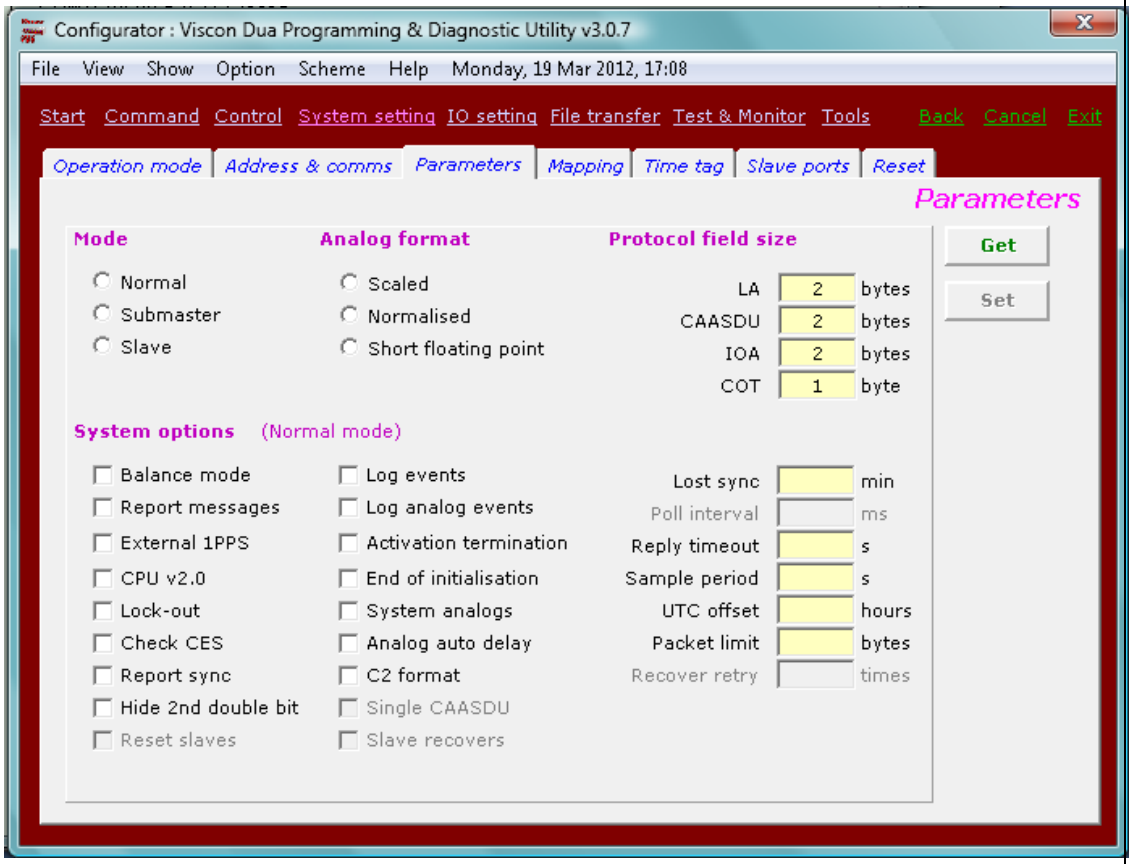
Configurator window: Address and Comms

- Protocol Address: as per setting in Comms. window
- Communication ports: must same as Serial port setting in Comms. window (Configuration Serial 1)



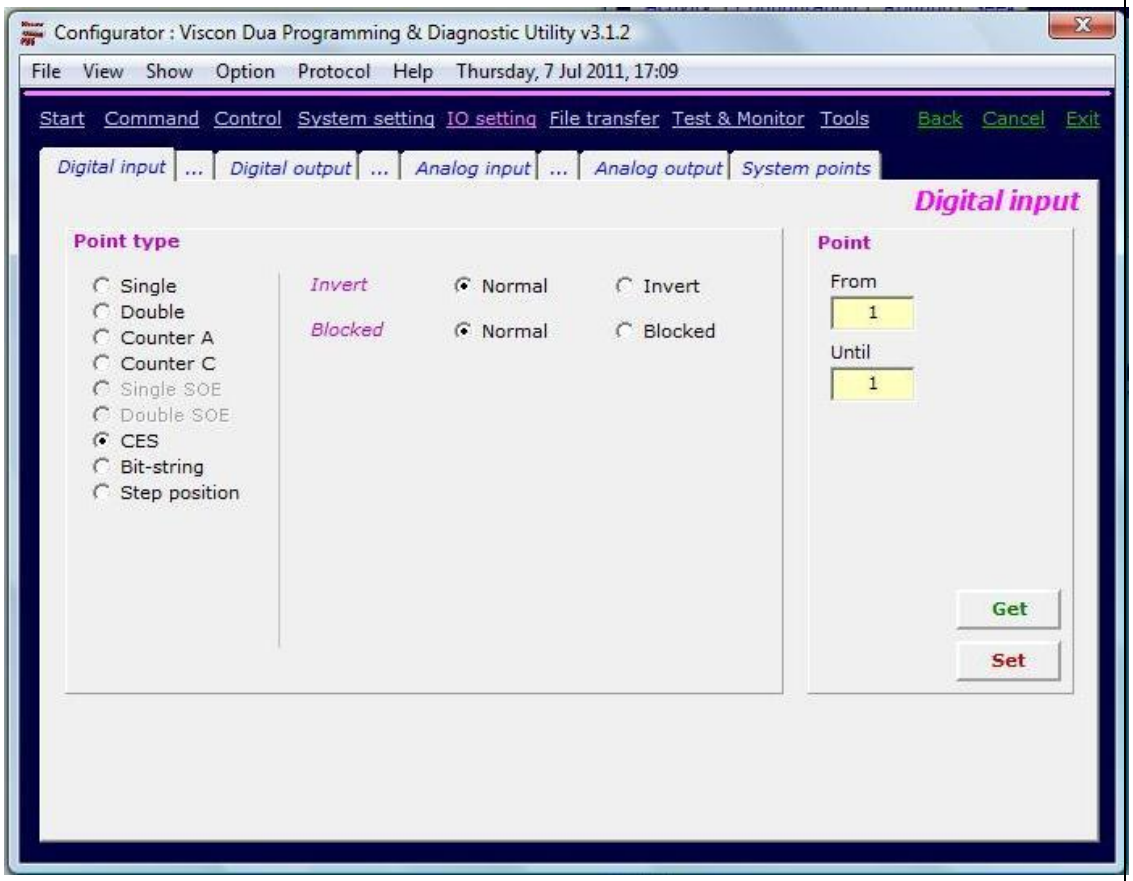
Configurator window: Parameters

- Mode:
  - **Normal:** RTU I/O reports direct to HX/MS (used for Configuration 2 and Configuration 4).
  - **Sub-master**: RTU I/O reports to Port 0 (Configuration 1 and Configuration 3).
  - **Slave**: Slave RTU I/O reports to sub-master (Configuration 1 and Configuration 3).
- Analog format:
  - **Scaled:** digital raw value from -4096 to +4095.
  - **Normalised**: 0 to +32767.
  - **Short floating point**: string value
- Protocol filed size: setting is based on MS

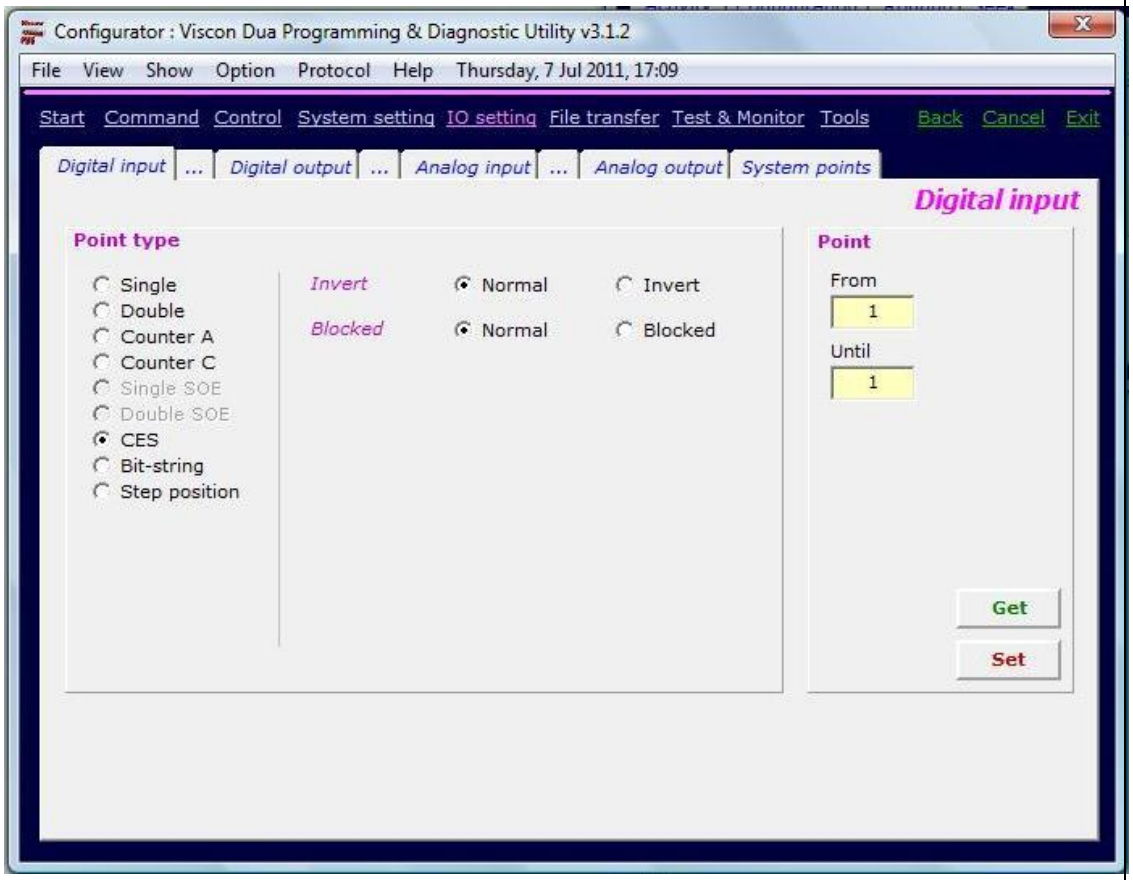


Configurator window: IO Setting for Digital Input

- Point type:
  - **Single**: single point that represents 2 statuses
  - **Double**: 2 points that represent 4 statuses
  - **Counter A**: single point to detect continuous pulse
  - **Counter C**: 2 points to detect continuous pulse
  - **CES**: DI point to represent 32-bit string
  - **Step position**: DI point to represent status of step position
  - **Invert**: set to Normal to receive actual value or set to **Invert** to receive opposite value
  - **Blocked**: set to **Normal** to receive status value or set to **Blocked** to ignore the status value

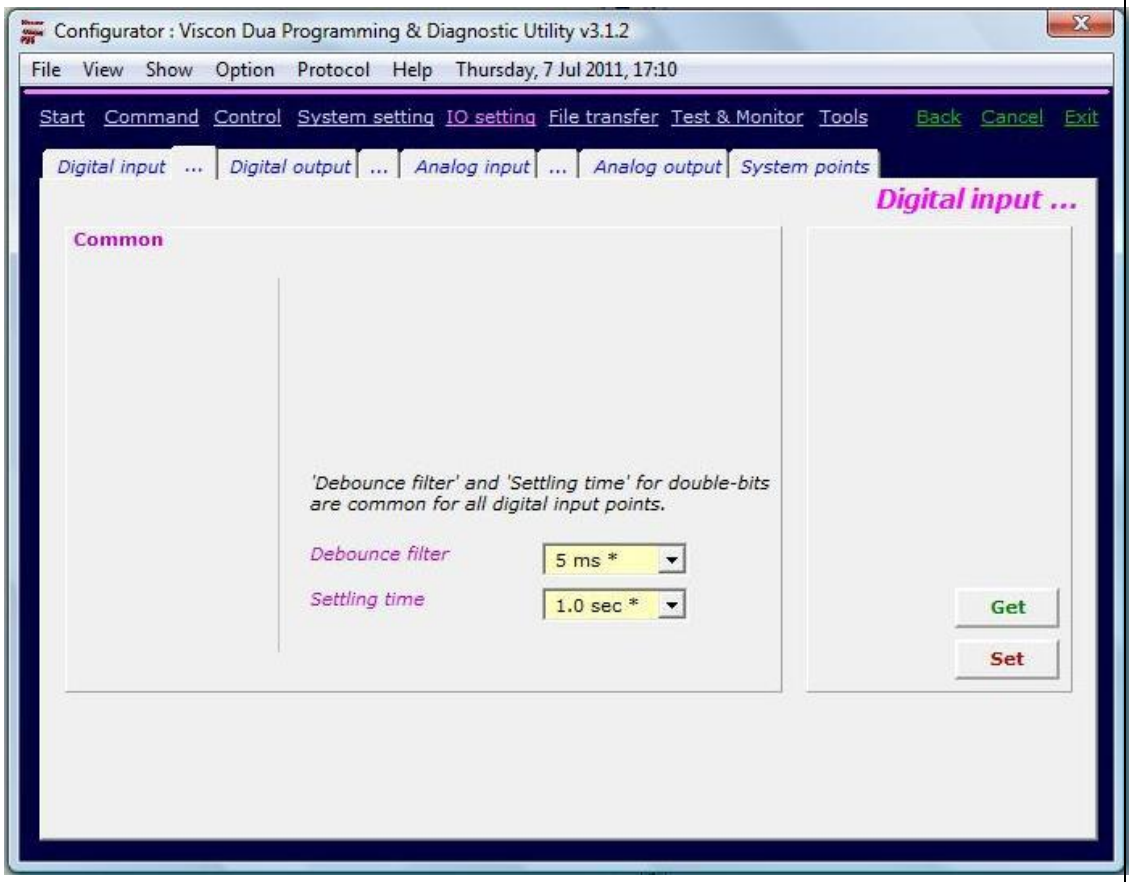


- **Point**: key in point number to receive the setting. Maximum range is 1 to 32.
- Click **Set** button to send setting to RTU.
- Click **Get** button to get setting from RTU.



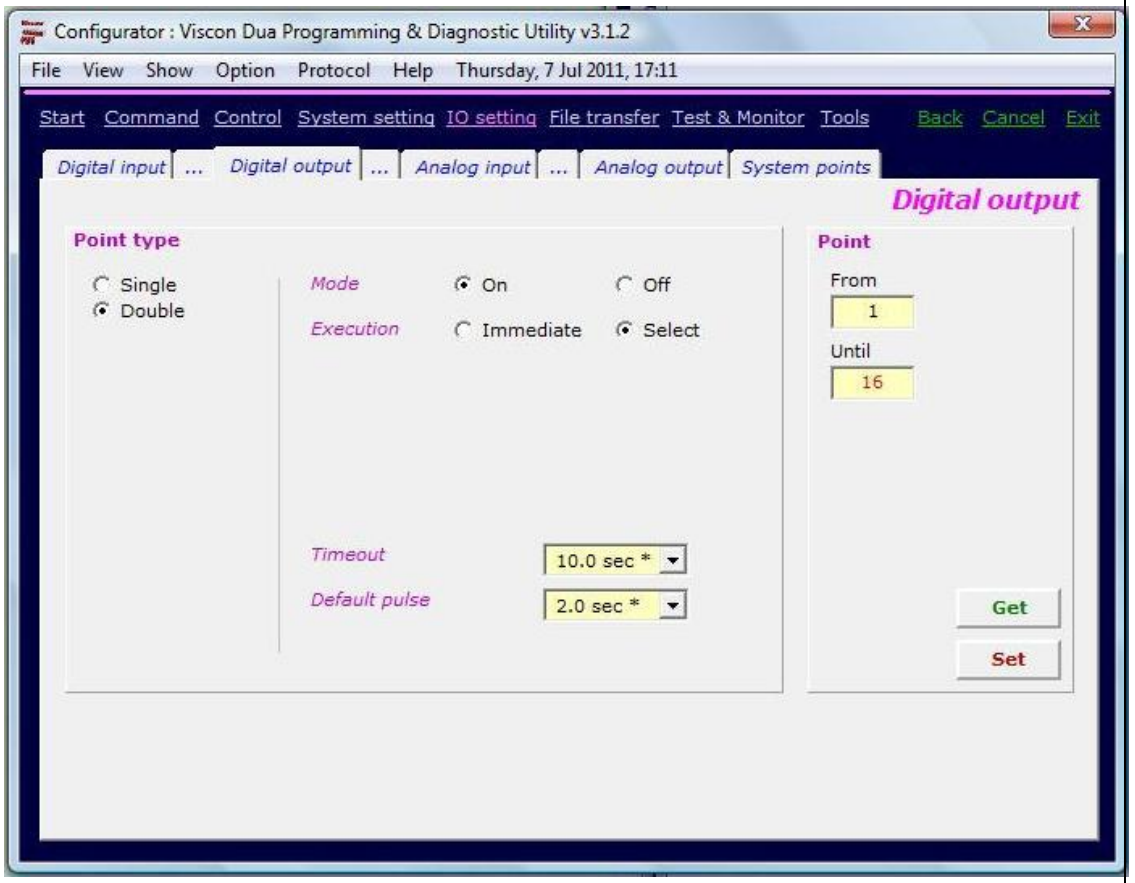
Configurator window: IO Setting for Digital Input

- **Debounce filter**: to ignore changes of double bit status within time
- **Settling time**: to accept double bit status as final status after settling time



Configurator window: IO Setting for Digital Output

- Point type:
  - **Single**: single point to control equipment with 2 status
  - **Double**: 2 points to control equipments with 4 statuses
- Mode:
  - **On**: send 'high' control command
  - **Off**: send 'low' control command
- Execution:
  - **Immediate**: require single command to execute control
  - **Select**: require '**Arm'** command before execute control
- **Timeout**: countdown between **Arm** command to **Execute**. **Arm** command will be ignored if no **Execute** command receive until timeout expire.
- **Default pulse**: valid duration of the relay close during execution of the control command
- **Point**: key in point number to receive the setting. Maximum range is 1 to 16.



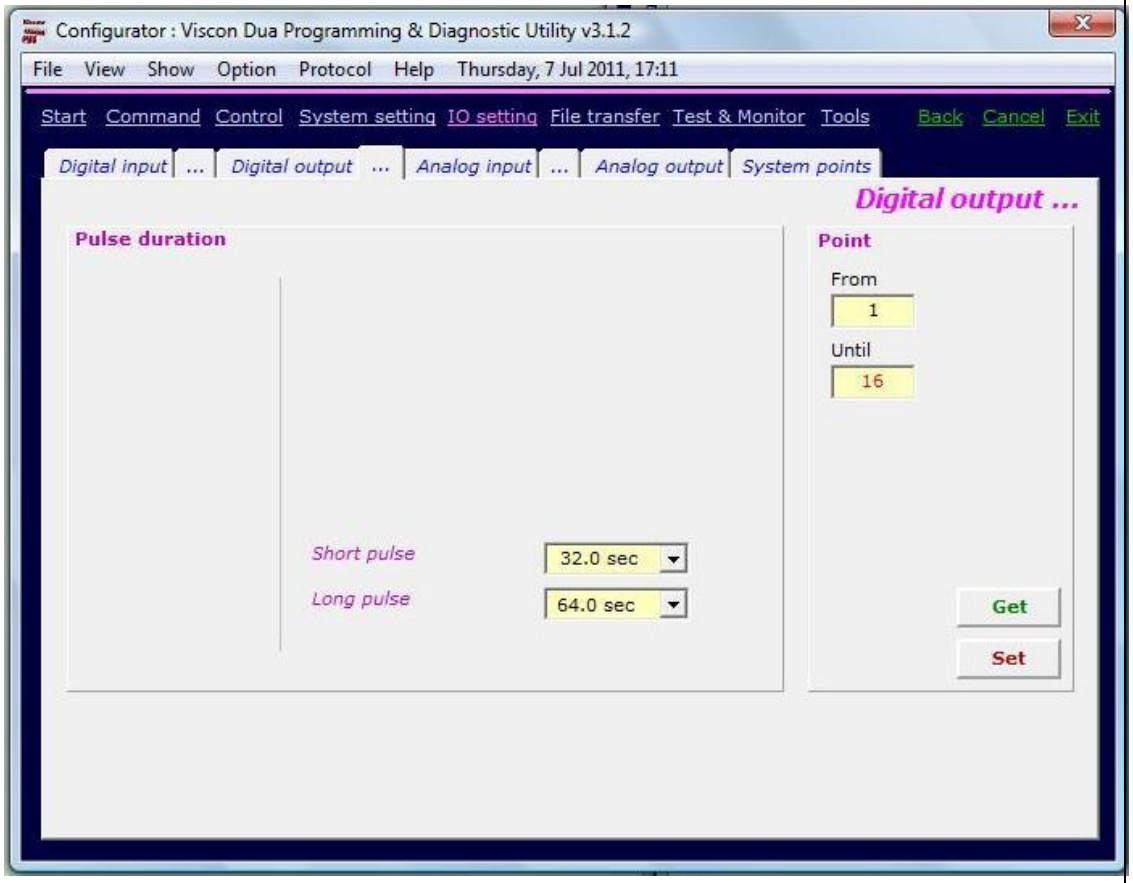
Configurator window: Digital Setting for Digital Input

- **Short pulse**: minimum valid pulse **Long pulse:** maximum valid pulse

- **Point**: key in point number to receive the setting. Maximum range is 1 to 16.

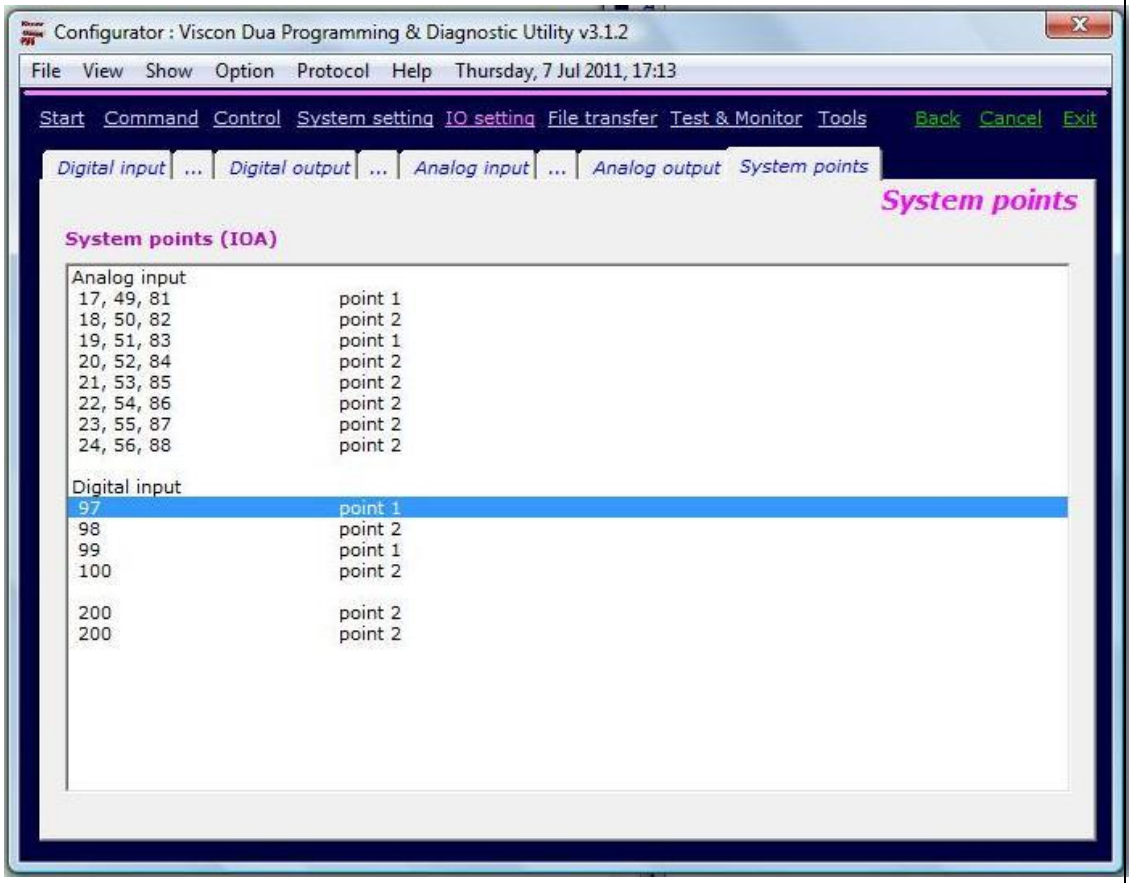
- Click **Set** button to send setting to RTU.

- Click **Get** button to get setting from RTU.



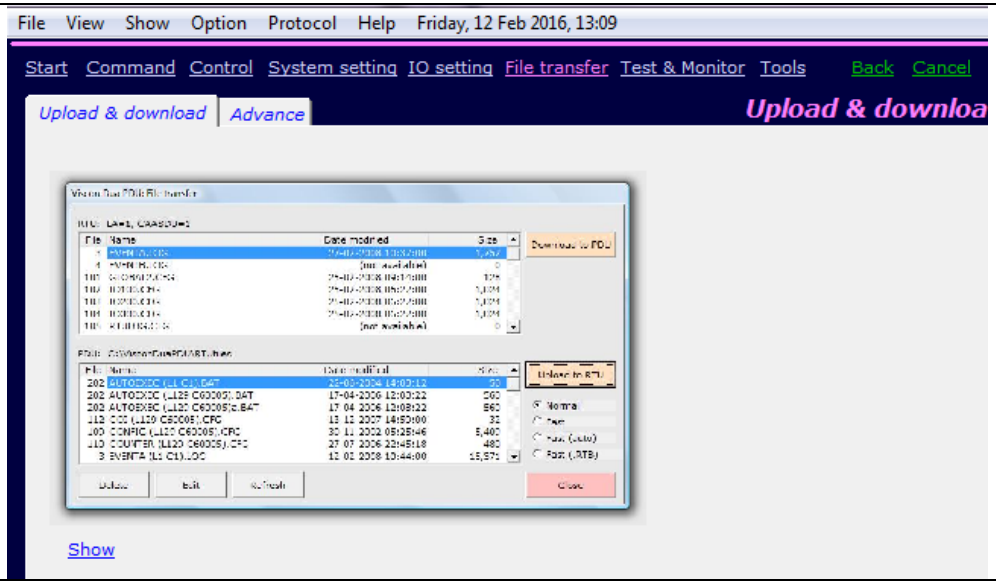
Configurator window: IO Setting System Point

- IOA = Information Object Address/Point address

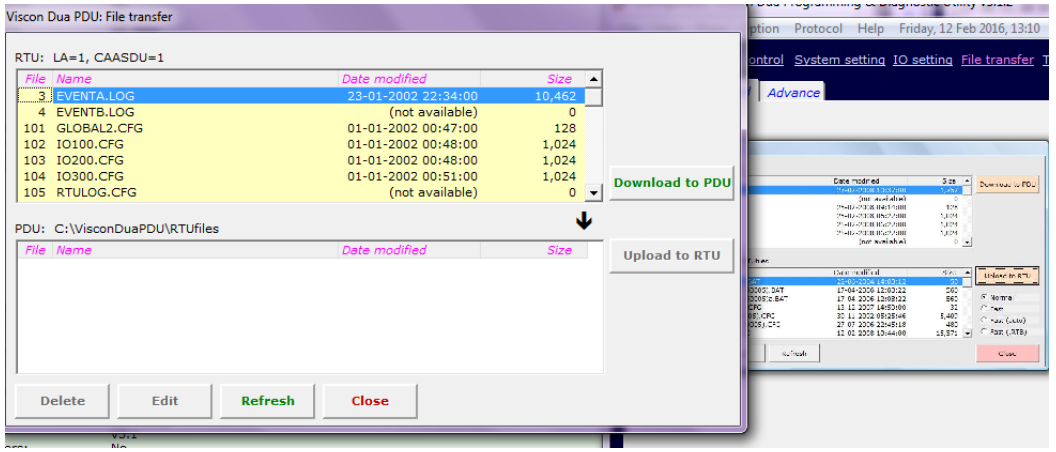


RTU Configuration: Download SOE buffer

- Select *File Transfer* menu

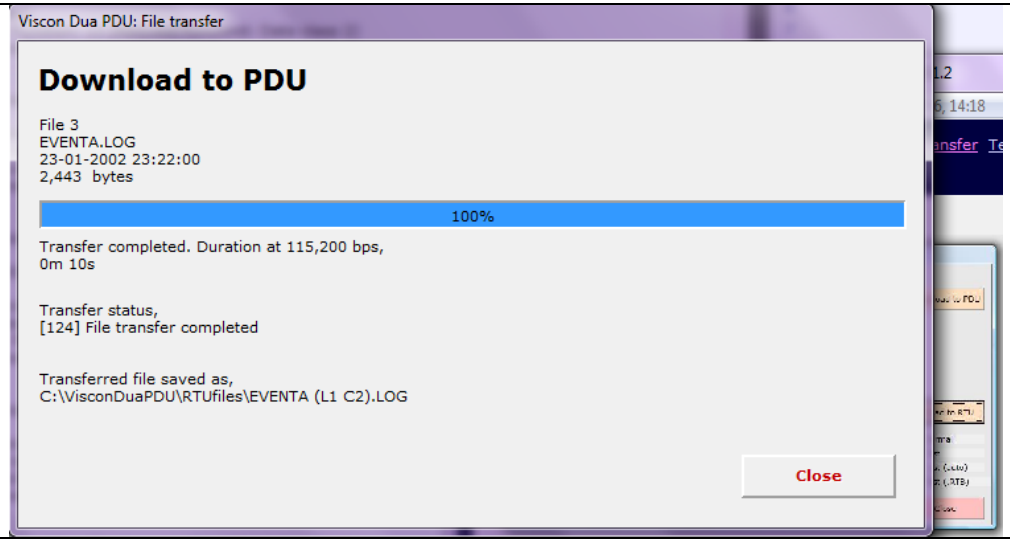


- SOE are configured to save in EVENTA LOG (file 1) which contain maximum 256 events.
- Click download to PDU button, PDU will install the log events to specific location.

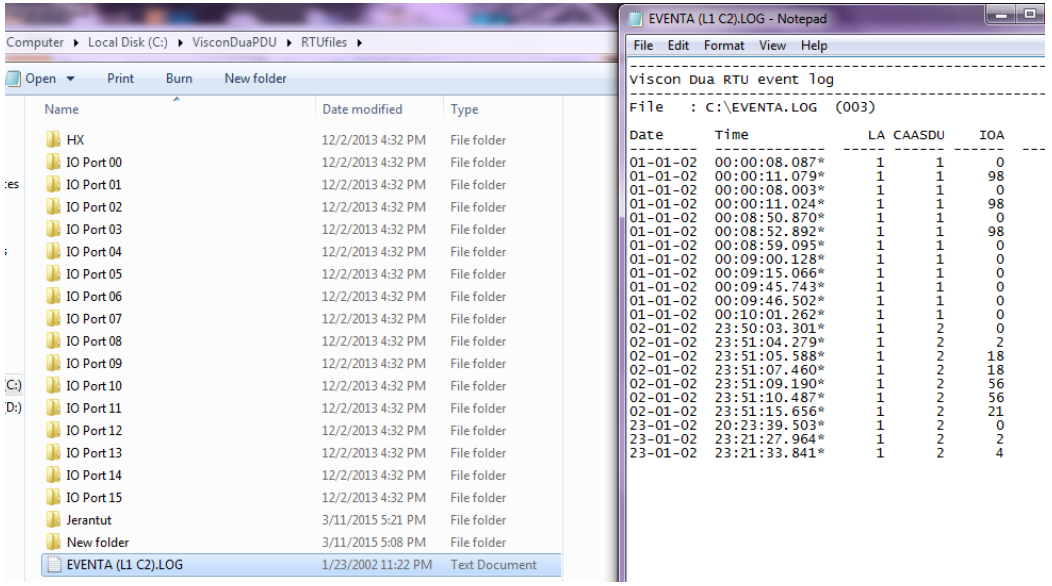


RTU Configuration: Download SOE buffer

- Log file is downloading.



- By default Event log shall be saved at Viscon Dua PDU folder in txt format.
- User can check at *Local disk (c:) / Viscon Dua PDU / RTU files/ Event (L1 C2).log*



-END OF MODULES-